Breathing Pattern Detection Using a Graphite-Based Paper Sensor for Development of an Auto-Calibrating CPAP System

Raunak Das

Breathing Pattern Detection Using a Graphite-Based Paper Sensor for Development of an Auto-Calibrating CPAP System

Submitted in partial fulfilment of the requirement for the award of the degree of

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in

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By:

Name	**University Reg. No**	**University Roll No**
Raunak Das	221260110033 (2022–26)	12622005033

Under the Supervision of

Prof. (Dr.) Madhurima Chattopadhyay
Head of the Department

Department of Applied Electronics & Instrumentation Engineering
Heritage Institute of Technology, Kolkata

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Prof. (Dr.) Madhurima Chattopadhyay
Head of the Department

Department of Applied Electronics & Instrumentation Engineering Heritage Institute of Technology, Kolkata

DEPARTMENT OF APPLIED ELECTRONICS AND INSTRUMENTATION ENGINEERING

HERITAGE INSTITUTE OF TECHNOLOGY

CERTIFICATE OF APPROVAL

This is to certify that the project work entitled

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has been successfully completed by

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Prof. (Dr.) Madhurima Chattopadhyay
Project Guide and Head of the Dept
Dept. of AEIE
.

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External Examiner

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External Examiner

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Raunak Das

Department of Applied Electronics & Instrumentation Engineering
Heritage Institute of Technology, Kolkata
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CONTENTS

Abstract

This research focuses on the design of an indigenous low cost auto regulating CPAP device using a biodegradable paper based humidity sensor with graphite. Experimental comparison between different paper substrate types such as Whatman Grade 1 filter paper, handmade paper, and polypropylene cloth found out that the best combination was achieved using the Whatman Grade 1 filter paper coupled with the use of thermal recovery heat pad which displayed highest humidity sensitivity, lowest baseline drift of −0.0976% within 10 cycles, and stable desorption sensitivity of −6.0339 ADC/%RH. The thermal recovery function enabled prevention of sensor saturation as well as increasing its longevity by at least twice from a single cycle to about three cycles. The designed system utilized ESP32 based embedded system, Bluetooth connectivity, OLED display, Android application and 3D printed BLDC pump with maximum calibrated CPAP pressure of 19.24 cmH_2O. These recordings and measurements helped prove the efficacy of real-time detection of abnormal respiratory activity, which when compared to literature, including increased rates of breathing and prolonging exhalation, are characteristics of obstructive airway disorders. Thus proving that a sustainable and inexpensive solution can be provided in a fully localized manner through an adaptive CPAP system.

Index Terms

Paper-Based Humidity Sensor, Auto-Regulating CPAP System, Graphite Sensor, Respiratory Monitoring, Sustainable Biomedical Device

I. INTRODUCTION

A. Introduction

As the prevalence of respiratory diseases increases globally, the need for an effective, continuous and easily accessible means of respiratory support is critical. CPAP, which is the standard of care for ailments such as obstructive sleep apnea, acute respiratory distress syndrome amongst others is highly efficacious in its treatment. However, due to the high cost, structural rigidity and proprietary sensing architecture, conventional CPAP units have failed to become accessible to many individuals across various settings.

In order to fill this technological gap, this research project proposes the design of a novel, auto-calibrating CPAP system utilizing an indigenous biodegradable paper humidity sensor based on biodegradable graphite. By capitalizing on the natural moisture dynamics present within physiological breathing, this custom designed sensor presents a highly sensitive non-invasive platform for respiratory tracking. This project presents the comprehensive methodology from sustainable material capture of physiological data, through customized signal acquisition through embedded electronics to control of an indigenous 3D printed brushless DC motor (BLDC) pump.

B. Respiratory Disorders, CPAP Systems, and Humidity Based Monitoring

There is a rising trend in the incidence of different respiratory diseases like COPD, OSA, Asthma, and other breathing disorders that have made these ailments global concerns within modern health care [1], [2]. The underlying problems with patients of such diseases include the inability to breathe properly, inefficiency in gas exchange, breathing instability, respiratory cessation while sleeping, and the need for constant monitoring and assistance in the breathing process.

The list of current technological solutions that can be used to help patients in cases like those includes Continuous Positive Airway Pressure (CPAP) systems, which can be considered one of the most used non-invasive techniques for treating respiratory problems [3], [4]. CPAP units work with the help of the provided flow of pressurized air that is delivered through the mask in order to keep air passages open and ensure efficient breathing in sleep and breathing assistance modes. Manufacturers such as ResMed and Philips Respironics create units with turbines, humidifiers, sensors, and electronic controllers for regulating the process of delivering air flow [5], [6]. However, CPAP systems are generally very expensive and rely on foreign equipment for operation.

Recent advancements in wearable healthcare systems and intelligent sensing technologies have encouraged the development of adaptive respiratory monitoring systems capable of continuously tracking patient breathing activity in real time [7]. Among various respiratory sensing approaches, humidity-based monitoring has gained significant attention because exhaled human breath contains substantially higher moisture concentration compared to ambient air. Consequently, the cyclic humidity variations produced during inhalation and exhalation provide a non-invasive mechanism for detecting respiratory activity, breathing rate, and breathing intensity [8]. Humidity-responsive sensors therefore, provide an effective foundation for wearable respiratory monitoring and adaptive CPAP regulation systems.

C. Types of Humidity Sensing Techniques

There exist different kinds of humidity sensing technologies depending on how they operate; these include capacitive, resistive, thermal conductivity, and optical [9]. Capacitive humidity sensing works on the basis that alterations in dielectric properties due to the absorption of water molecules within the hygroscopic substance placed between the electrodes are converted to signals. It is energy-efficient, and hence, highly reliable in giving accurate results in industries and the environment, among other sectors [10].

On the other hand, resistive humidity sensing works on the basis of alterations in the electrical resistance of moisture-dependent compounds as a result of the ionization or electron conduction within polymers, salt solutions, ceramics, or carbon conductors [10]. Due to their simple construction, ease of manufacture at a low cost, and applicability on flexible substrates, they can be considered for use as disposable wearables.Humidity sensors based on thermal conductivity take advantage of variations in heat conduction due to differences in dry and humid air environments.

Although they can be used for the measurement of absolute humidity, they are characterized by relatively high levels of energy usage and large sensor sizes [11]. Optical humidity sensors exploit modifications in the optical characteristics (e.g., reflection, scattering, absorption, etc.) caused by different humidity levels and have high-precision sensing capability, although at the same time they necessitate rather complicated and costly hardware equipment [12]. Among other types of sensors, conductive carbon-based resistive humidity sensors have been attracting more attention due to several distinctive features.

D. Limitations of Conventional Humidity Sensors and the Shift Toward Sustainable Sensing

Despite their widespread usage, conventional humidity sensors suffer from several limitations associated with fabrication cost, environmental sustainability, mechanical rigidity, and long-term operational stability [13], [14]. Many commercial sensors rely on expensive materials, complex fabrication procedures, and non-biodegradable electronic components that contribute significantly to electronic waste generation and environmental burden [15], [16].

The conventional method of humidity sensing suffers from being incompatible with wearable health monitoring due to inflexible construction and lack of mechanical flexibility. Moreover, prolonged exposure to humidity may cause problems such as sensor drift, sensor contamination, hysteresis, and loss of accuracy. [9]. Thermal and optical humidity sensors may additionally exhibit high power consumption, making them unsuitable for portable battery-powered respiratory monitoring systems.

To overcome these limitations, recent research has focused more and more on sustainable and biodegradable sensing technologies based on flexible materials such as cellulose paper, textile substrates and conductive carbon-based structures [15], [16]. The paper-based humidity sensors have advantages, such as low cost of manufacture, biodegradability, lightweight structure, mechanical flexibility, and compatibility with wearable healthcare systems. In particular, graphite based paper sensors provide a simple and scalable sensing architecture where the moisture absorption induced by humidity modulates ionic and electronic conduction pathways resulting in measurable resistance variation [17].

Motivated by these advantages, the present work investigates multiple biodegradable and flexible substrate materials, including Whatman Grade 1 filter paper, handmade paper, and polypropylene cloth-based sensing structures using graphite and sodium chloride conductive architectures. Their sensing performance is systematically evaluated under controlled humidity cycling conditions to analyse sensitivity, repeatability, degradation behaviour, and operational stability during simulated respiratory conditions.

E. Sustainable Development Goals and Indigenous Healthcare Technology

The evolution of sustainable wearables for healthcare applications goes hand-in-hand with several United Nations Sustainable Development Goals (SDGs), including those concerning responsible consumption and production, good health and well-being, and sustainable industry innovations [15], [16]. The use of biodegradable paper-based sensing systems allows decreasing the need for non-renewable components and reduces e-waste produced, making the technology development environmentally friendly.

On top of that, along with sustainability, affordability and availability are other pressing problems facing modern respiratory healthcare systems, especially when it comes to developing countries. Traditional commercial CPAP systems involve importing expensive airflow modules, electronics, and spare parts, which makes the systems unaffordable for most patients. Therefore, the development of indigenous low-cost respiratory support systems represents an important technological and socioeconomic objective.

Motivated by these challenges, the present work extends beyond standalone humidity sensing toward the realization of a fully indigenous wearable respiratory monitoring and autoregulating CPAP platform. The developed system integrates an optimised humidity sensor, wearable nebulizer mask architecture, thermal stabilization subsystem, custom PCB, ESP32-based embedded controller, Android application interface, and an in-house fabricated BLDC turbo impeller pump manufactured using additive manufacturing techniques.

The resulting platform forms a closed-loop respiratory assistance system in which breathing-induced humidity variations dynamically regulate airflow delivery through adaptive pump control. Therefore, this work represents not only the development of a biodegradable humidity sensor but also the realization of a compact wearable respiratory monitoring and support device aligned with sustainable healthcare technology and the broader objectives of the “Make in India” initiative.

F. Market Study

The current state of the global market for continuous positive airway pressure (CPAP) devices is characterized by a dominant presence of just a few multinational companies producing these systems, such as ResMed and Philips Respironics, with their widespread use in clinics and for home-based respiratory therapies [5], [6]. Commercial CPAP platforms include complex airflow generation devices, pressure control systems, humidification modules, respiratory monitoring electronics, and embedded control systems allowing the provision of continuous airway pressure for patients diagnosed with obstructive sleep apnea and chronic respiratory problems.

A commercial survey conducted within the territory of India has shown that these imported respiratory assistance systems still pose a relatively high price for the majority of the population. Products like the ResMed AirSense 11 AutoSet CPAP machine or the Philips Respironics DreamStation Auto CPAP machine can be purchased at a price ranging between approximately INR 1,00,000 to 3,80,000. Apart from the basic device cost, additional costs related to replacement components like blower motor assembly, humidifiers, turbine assembly, masks, and accessories also make up for higher costs involved in long term operations. Commercially available replacement blower motor assemblies could be priced as high as around INR 20,000 or more in the consumer market [18]. Commercial pricing trends and blower motor replacement costs related to imported CPAP devices are shown in Fig. 1.

In addition to the above costs, the prevailing market scenario shows significant dependency on imported medical devices and proprietary manufacturing processes. Critical sub-assemblies used in the design of these machines include blower turbos, miniature centrifugal turbines, brushless DC motors, pressure regulators, and medical sensors, which are mainly available

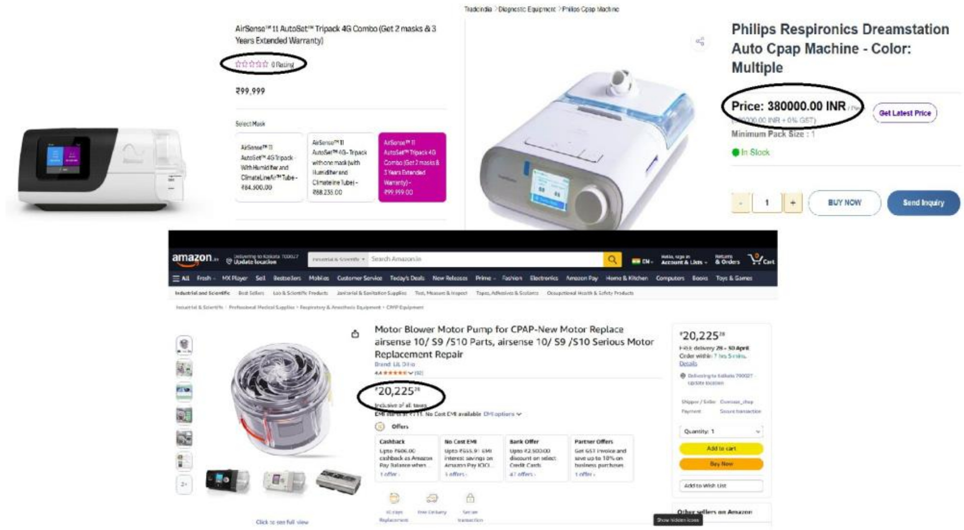


Fig. 1: Available commercial CPAP devices and blower motor assemblies in online B2C marketplaces [19].

through international distribution channels [20]. This makes these devices expensive in terms of device cost, accessory costs, and maintenance capabilities.

Also, many commercially available CPAP devices work in proprietary hardware environments where firmware manipulation, interchangeable parts, and repair facilities are under the control of the manufacturer. This trend creates a high degree of reliance on maintenance while making it hard to perform affordable repairs and customizations. The mentioned tendencies demonstrate a strong demand for cost-effective respiratory devices that can decrease economic burdens related to prolonged CPAP treatment [21].

II. Device Fabrication

This project's process flow will follow four different stages, ranging from the basic examination of materials to system level integration and calibration to fit human physiology. The primary objective of the project involves the implementation of a specific humidity sensing interface to enable a respiratory adaptation control system.

The study starts with the initial stage, which involves setting up the human to sensor interface through comparative analysis between various handmade humidity sensors made from graphite to determine the best choice among the sensors for tracking human respiration. After identifying the sensor to be used, the next step involves data processing, which explains the necessary processing techniques required to change the input data to stable form through use of different algorithms. After processing, the data is used in the next stage to facilitate physical actuation through the control mechanism involved in running the brushless DC (BLDC) pump motor. The last step involves system integration where all the systems are put together to achieve a functional Continuous Positive Airway Pressure (CPAP) machine capable of operating automatically with an ideal rate of human breathing.

III. Experiments for Sensor Development

A. Paper as an alternate medium

Paper, due to its biodegradability, abundance, and low cost, presents an ideal substrate for environmentally sustainable sensor technology [22], [23]. Utilizing paper substrates significantly reduces reliance on non-renewable resources [22]. Additionally, the manufacturing process is simple. Using common materials like pencils and copper tape eliminates the need for complex, energy-intensive industrial processes, making production both facile and cost-effective [24].

The choice of paper type is crucial before sensor fabrication since substrate properties directly influence device performance [22]. Various types of paper have been compared for electrode fabrication, including filter paper, vegetable parchment, office paper, photo paper, and chromatography paper [24], [25]. Whatman Grade 1 filter paper is widely recognized as a robust substrate choice for sensors due to its consistency, purity, and physical properties. With medium retention (11 µm) and flow rate, Whatman Grade 1 is suited for applications that demand both mechanical stability and accessibility for electrode fabrication [26]. Its composition of high-quality alpha-cellulose ensures minimal impurities and high wet strength, which is valuable in sensor construction where durability and uniformity are critical [22].

Due to these properties, Whatman Grade 1 is frequently chosen for fabricating electrodes and as a versatile substrate in paper-based sensors, especially when reliable performance and simple processing are priorities [24].

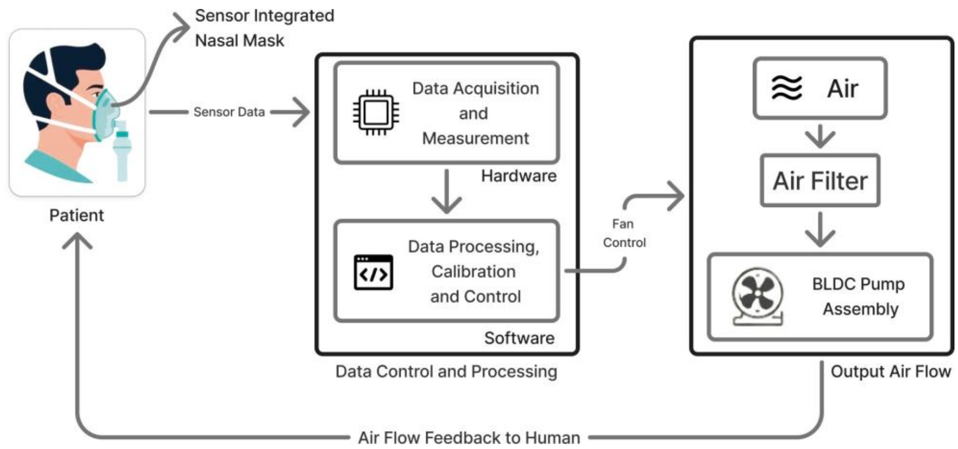


Fig. 2: Block Diagram

Guided by this evidence, sensor development and testing in this work focused on two types of paper substrates.

B. Electrode Selection

In this study, electrodes were fabricated using 10B-grade graphite pencils directly on paper substrates due to their ease of use, low cost, and accessibility [27]–[29].

Graphite pencils having 10B composition ensure conductivity for sensing, as initial resistances of such sensors vary between 40 kΩ to 70 kΩ, which makes them compatible with the usage with voltage divider and analogue to digital conversion systems [27], [29].

Use of graphite pencil electrodes allows easy prototyping without the need for any additional equipment and material. [28]. However, experimental results revealed challenges related to electrode durability. The graphite traces degrade over time, mainly due to moisture absorption and physical wear during repeated humidity cycles [29]. This degradation is evident from significant increases in resistance, sometimes exceeding 1 MΩ, leading to signal inaccuracies and sensor failure [28], [29].

While the use of graphite pencils enables functional sensor prototypes, limitations in long-term stability indicate that alternative electrode materials such as metallic or graphene inks may improve performance [30]–[32]. These alternatives provide enhanced conductivity, better adhesion, and greater resistance to moisture-induced degradation, which can extend sensor lifespan and reliability in demanding applications [30], [31].

Therefore, the electrode choice reflects a balance between fabrication simplicity and sensor durability. Graphite pencils serve well for initial testing and exploration, but more advanced materials are recommended for future development to address stability concerns [28], [30]–[32].

C. Graphite vs Silver Chloride

The electrode material most widely used in electrochemical detection systems is $AgCl$. In this study, however, the material chosen as the primary electrode is graphite, due to the excellent stability, elasticity, and durability of graphite as an electrode under humid conditions. While $AgCl$ has been widely used as a reference electrode material for its electrochemical stability, $AgCl$ has several practical drawbacks for use as the primary sensing electrode in wearable humidity sensing applications [33], [34].

A significant limitation associated with the use of the $Ag/AgCl$ system lies in the dependency of the equilibrium of the electrode on chloride ions. The electrode potential is given by:

$$E = E^{\circ} - \frac{RT}{F} \ln a_{Cl^-}$$

Therefore, the variations of chloride ions concentration at the local level will significantly impact the electrode potential and lead to the occurrence of baseline drift during sensing [35]. Moreover, silver chloride exhibits the phenomenon of photoreduction when continuously exposed to visible light, thereby leading to its gradual transformation into silver [36]. Apart from that, one of the most serious problems related to the usage of $AgCl$ is connected to its mechanical behavior. Indeed, after its deposition and drying, this material becomes relatively hard and brittle. This means that, since the proposed biosensor platform should have flexibility and can be repeatedly bent, cracking in the $AgCl$ layer will occur, which will lead to unstable electrical behavior of this structure.

Moreover, apart from all mentioned above, the usage of $AgCl$ as a sensor material brings other issues associated with its conductive nature. Namely, due to its low resistance, silver-based sensors demonstrate high conductivity, which can alter the sensing principle of the whole system because of the required changes in the resistance values.

On the other hand, graphite acts as an environmentally stable and structurally flexible sp^2 hybrid carbon structure that is resistant to corrosion and fouling by chemicals [37]. Contrarily to $AgCl$, graphite electrode retains its integrity even when bent or strained repeatedly; therefore, this material becomes appropriate for use in flexible and wearables sensors. Graphite film preserves structural stability during deposition and will not crack under bending strain and integration with mask. Besides, graphite electrode has the advantage of resistive behavior which can be exploited for humidity sensing without affecting the whole sensing architecture.

There are many benefits related to the implementation of graphite electrodes including affordable manufacturing process, simple deposition, suitability for paper substrate, and no need for specialized storage [38]. Such properties render graphite extremely convenient for designing economical and environmentally friendly respiratory masks fabricated under conditions of limited resources. Comparison between graphite and silver chloride electrode is provided in Table I.

TABLE I: Comparison Between Graphite and Silver Chloride ($AgCl$) Electrodes

Evaluation Metric	**Graphite**	**Silver Chloride (*AgCl*)**
Electrochemical Behaviour	Provides stable resistive conduction suitable for humidity sensing.	Acts as a reference type electrochemical material dependent on ionic equilibrium.
Chemical & Light Stability	Chemically stable and unaffected by ambient light exposure.	Prone to photoreduction and gradual surface degradation under light exposure.
Mechanical Flexibility	Flexible and maintains integrity during repeated bending and mask integration.	Becomes rigid after drying and develops cracks during bending or flexing.
Humidity Sensing Compatibility	Maintains localized resistive variation required for sensing operation.	Highly conductive behaviour can alter the intended sensing mechanism.
Measurement Stability	Compatible with simple resistive measurement circuitry.	Requires more complex electrochemical measurement and stabilization methods.
Fabrication & Cost	Low cost, easily deposited, and compatible with paper substrates.	Relatively expensive and requires controlled deposition conditions.
Operational Suitability	Suitable for wearable and flexible respiratory sensing systems.	Less suitable for flexible wearable applications under repeated deformation.

D. Development and experimental testing of paper sensor

In this section, we have fabricated the sensor, and the detailed fabrication technique is described.

1) Sensor Devlopment:

- **Material selection:**

Grade A filter paper, with a thickness ranging from 0.17 mm to 0.93 mm, was selected as the substrate for the flex sensor due to its lightweight, flexible, and porous structure. A 10B high-quality graphite pencil was used to coat the paper evenly, forming a conductive electrode. The thin copper tape was applied to both edges of the sensor, serving as a probe to facilitate electrical connections. Additionally, distilled water and sodium chloride (NaCl), commonly available as table salt, were employed during the fabrication process to enhance conductivity.

- **Chemical Properties**

The chemical properties of graphite and sodium chloride (NaCl) are integral to the sensor's enhanced performance:

(a) Graphite: Graphite, a crystalline form of carbon, is characterized by its excellent electrical conductivity, approximately 3×10^3 S/m. Its honeycomb-like layered structure consists of hexagonally arranged carbon atoms, where delocalized electrons move freely between layers, enabling efficient electrical conduction. This layered structure also imparts flexibility, making graphite suitable for applications requiring mechanical deformation, such as flex sensors [39], [40], [41], [42].

(b) Sodium Chloride (NaCl): When dissolved in distilled water, NaCl dissociates into sodium (Na^+) and chloride (Cl^-) ions, as represented by the dissociation equation:

$$NaCl\ (s) \xrightarrow{H_2O} Na^+(aq) + Cl^-(aq)$$

These ions enhance the sensor's conductivity through ionic conduction, which complements the electronic conduction provided by graphite. This synergistic effect significantly improves the sensor's sensitivity and performance.

This process generates sodium (Na^+) and chloride (Cl^-) ions, enhancing the graphite-coated paper's electrical conductivity. These ions stabilise the electrical pathways, improving the sensor's sensitivity and responsiveness to mechanical deformations. The combination of graphite's conductive, layered structure and NaCl's ionic dissociation significantly enhances the sensor's overall performance across various applications [43].

- **Geometry selection:**

Then the outline structure of the sensor is drawn on the filter paper and the area is filled by the pencil graphite as shown in fig.3

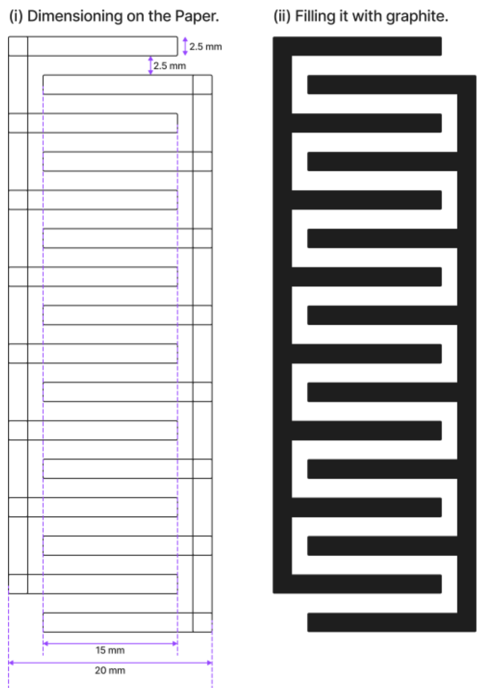


Fig. 3: Geometric dimensions of the sensor

- **Sensor fabrication:**

The sensor fabrication process begins by first outlining the sensor on a white filter paper, as shown in Fig.3. Step (i), Creating precise straight lines using a ruler as per the specified measurements: a total width of 20 mm, an internal pathway width of 2.5 mm, and a 2.5 mm gap between adjacent paths. This step ensured uniformity and accuracy in the interdigitated pattern. Once the outline is complete, the next step (ii) involves filling the designated area with graphite using a 10B graphite pencil. This step is crucial to creating a consistent, even layer of graphite, which serves as the primary conductive element of the sensor.

Step (iii) after applying the graphite layer, a layer of sodium chloride (NaCl) solution is applied using a dropper on the surface of the sensor where graphite is deposited, ensuring complete coverage as shown in Fig.7. This solution was prepared by dissolving 1 gram of common kitchen salt (NaCl) in 5 ml of distilled water (H_2O), achieving a solubility ratio of 1g/5ml. Then the paper is left to dry overnight under normal room conditions. This step enhances the electrical conductivity of the graphite by introducing ions from the NaCl solution, which improves the sensor's sensitivity.

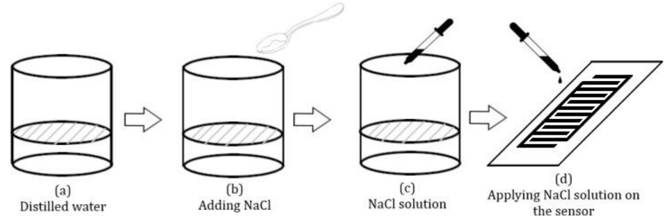


Fig. 4: Step (iii) (a-d) Schematic demonstration of NaCl solution preparation and application on the sensor.

Once the paper is completely dry, copper tape is placed on both sides of the graphite-coated area to act as electrical probes, as shown in fig.5 and step (iv).
In the final stage of fabrication (step (v)), two wires are soldered to the copper tape at each end of the sensor. These wires provide the necessary electrical connections, allowing the sensor to be integrated into external circuits for testing and use, as shown in Fig.7.

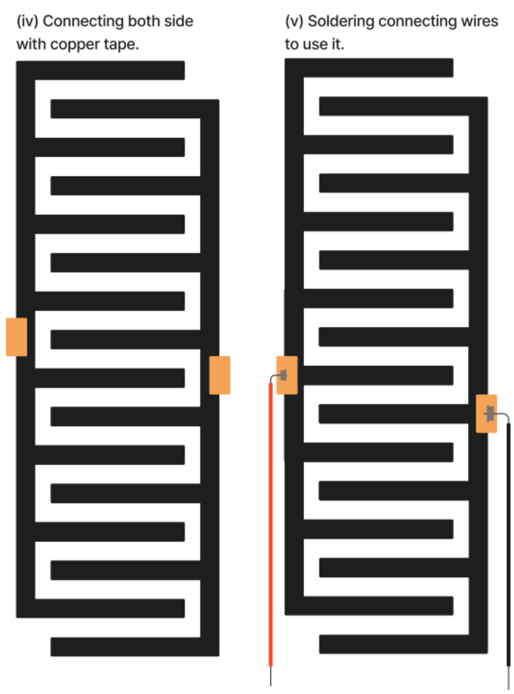


Fig. 5: The completely developed paper sensors.

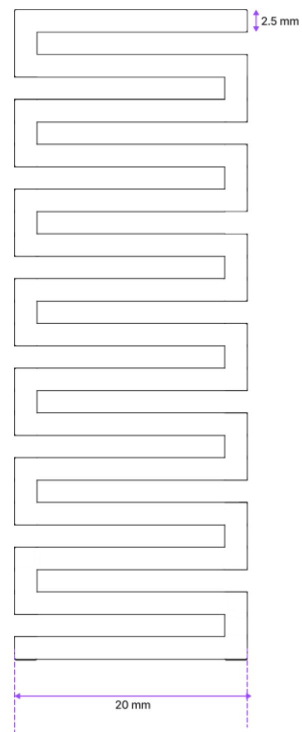


Fig. 6: The developed heatpad.

2) Heat Pad Fabrication: The heat pad used to maintain sensor dryness is fabricated from aluminium foil arranged in a serpentine pattern, as illustrated in Figure 6 [44]. When a voltage is applied across the foil, resistive heating occurs, raising the temperature of the surrounding substrates and helping to evaporate moisture accumulated during measurement cycles.

The serpentine pattern increases the effective length of the foil trace, providing distributed and uniform heating over the sensor area. The heat pad measures 20 mm in width and 2.5 mm in trace height, matching the sensor dimensions for optimal thermal contact and minimal energy loss. This integrated heating mechanism enables repeated sensor operation by mitigating moisture-induced degradation, as demonstrated in experimental runs [45].

E. Component List , Experimental Setup and Process

1) Controlled Humidity Chamber: A custom humidity-controlled chamber was constructed to simulate dynamic relative humidity (RH) conditions within a controlled environment. The chamber consisted of a cubical glass enclosure measuring 2 ft × 2 ft × 2 ft, designed to ensure consistent exposure of the sensors to varying humidity levels.

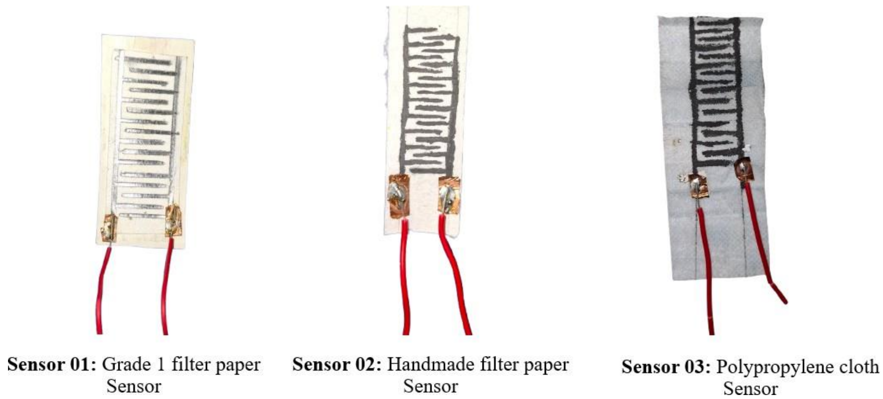


Fig. 7: The 3 types of sensors that were developed and analyzed for optimization

Two openings were positioned to optimize airflow and humidity distribution: an intake hole near the bottom on the front face to introduce humid air, and an exhaust hole near the top on the opposite back face for air expulsion. This offset configuration enhanced air circulation and promoted uniform humidity throughout the chamber.

Airflow was driven by two fans, one at the intake actively pulling humid air into the chamber, and one at the exhaust expelling air as RH neared saturation. Both fans were connected to relay modules controlled via an Arduino Uno R3, which dynamically regulated airflow based on real-time humidity readings to maintain set RH conditions.

2) Sensor Placement: The paper-based graphene humidity sensor and a commercial DHT11 humidity sensor were placed at the farthest point from the inlet inside the chamber. This ensured maximum and uniform exposure to humid air, minimised turbulence, and created stable and realistic conditions for sensor testing.

3) Fan Control Logic: Humidity inside the chamber was continuously monitored by the DHT11 sensor. When RH reached 90%, the exhaust fan was triggered through the relay, allowing for system delay and permitting RH to reach nearly 100%. Simultaneously, the external humidity source (e.g., water steam generator) was manually removed to reverse the humidity trend. This controlled up-and-down RH cycling generated clean and interpretable humidity response curves for sensor calibration.

4) Data Acquisition and Analysis: The paper-based sensor was integrated into a voltage divider circuit, with analogue voltage signals representing sensor resistance changes logged via the Arduino Uno R3. Data acquisition occurred during humidity increases from approximately 40% to 95% RH and decreases from 95% to 50% RH.

Collected data underwent post-processing, including noise reduction by averaging, and were plotted alongside the DHT11 sensor reference data. This process allowed for detailed analysis of the sensor's dynamic response characteristics and hysteresis under controlled, variable humidity environments.

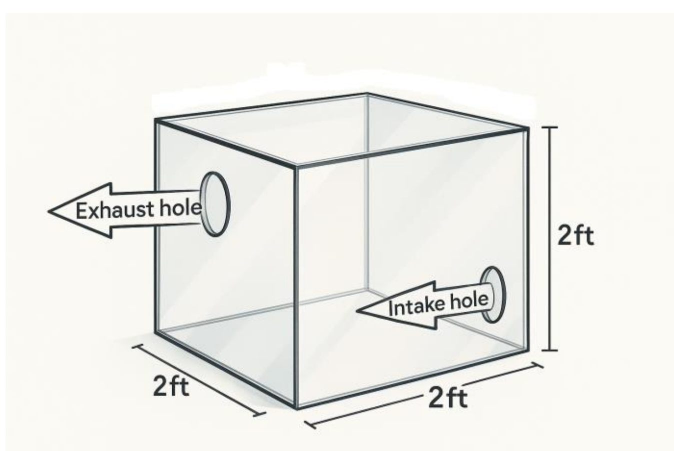


Fig. 8: Chamber dimensions and hole placements

5) Component List: The list of components used, along with their specifications, are mentioned in Table II.

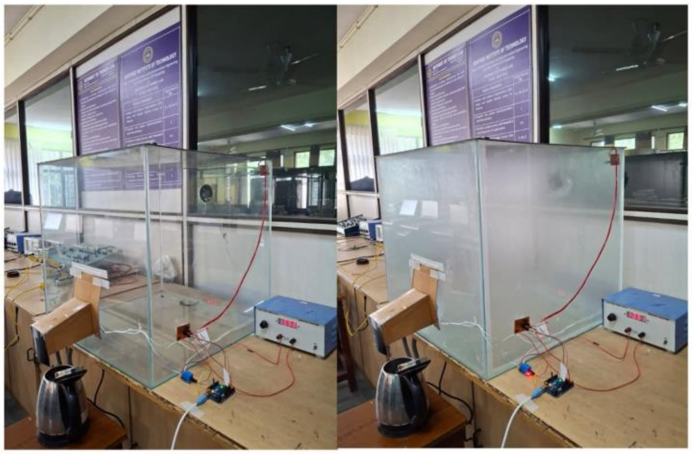


Fig. 9: Humidity at 40% (left) and 95% (right)

F. Calibration Calculations

The paper-based humidity sensor was calibrated by comparing its output against a DHT11 sensor [46]–[48]. The setup used a voltage divider circuit with the paper sensor and a known resistor R_{fixed}, and the voltage across the fixed resistor was read by the Arduino Uno's ADC, which features 10-bit resolution and a digital output range of 0–1023 [46].

1) Voltage Divider Principle:

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_{\text{fixed}}}{R_{\text{fixed}} + R_{\text{sensor}}} \tag{1}$$

Rearranged to solve for sensor resistance:

$$R_{\text{sensor}} = R_{\text{fixed}} \left(\frac{V_{\text{in}}}{V_{\text{out}}} - 1 \right) \tag{2}$$

Where:

- $V_{\text{in}} = 5.12$ V
- $V_{\text{out}} = \frac{\text{ADC value}}{1023} \times V_{\text{in}}$
- $R_{\text{fixed}} = 10$ kΩ

2) Example Calculation: Assuming at 80% RH:

- ADC value = 410
- $V_{\text{out}} = \frac{410}{1023} \times 5.12 \approx 2.05$ V

Then:

$$R_{\text{sensor}} = 10k \times \left(\frac{5.12}{2.05} - 1 \right)$$
$$\approx 10k \times (2.497 - 1)$$
$$\approx 14.97 \text{ k}\Omega$$

TABLE II: List of Components Used in the Breathing Rate Detection and CPAP Autoregulation Prototype

Component	**Specification**	**Purpose**
Whatman Grade 1 filter paper	0.17 to 0.93 mm thickness	Substrate for paper humidity sensor
Handmade paper sheets	Variable thickness high porosity	Comparative substrate for sensitivity testing
Polypropylene mask derived cloth sensor	Melt blown and spunbond polypropylene	Cloth based humidity sensor derived from standard mask material
Graphite pencil	10B high carbon content	Electrode fabrication on paper and polypropylene substrates
Copper tape	Adhesive conductive strip	Electrical contact and probe interface
Sodium chloride	1 g in 5 ml distilled water	Ionic conduction enhancer improving humidity response
Distilled water	Laboratory grade	Solvent for NaCl solution
Jumper wires	Male to female	Electrical connections for sensor interfacing
Soldering wire and iron	Lead-free solder	Permanent electrode to wire connection
Arduino Uno R3	ATmega328P microcontroller	Data acquisition and fan control in humidity chamber
10 kΩ resistor	±1 percent tolerance	Reference resistor for voltage divider calibration
Glass chamber	2 ft $\times$ 2 ft $\times$ 2 ft	Controlled humidity environment for sensor testing
DC fans	12 V axial type	Airflow regulation inside chamber
Relay modules	Single channel 5 V	Fan switching based on humidity threshold
Electric Water Heater (Kettle)	Portable type	Humidity generation for controlled cycling
Aluminum foil heat pad	Custom fabricated	Thermal stabilization of sensor during repeated cycles
Nebulizer mask	Medical grade PVC	Planned integration of humidity sensor for breathing analysis
CPAP device	Variable pressure type	Target system for autoregulated airflow
DHT11 Sensor Module	Temperature and Humidity Sensor with $\pm 2^{\circ}$C temperature accuracy, $\pm$5% RH humidity accuracy, and 3–5 years typical longevity	Measures ambient temperature and humidity for environmental monitoring and control
Power supply	5 V and 12 V regulated	Electronics and fan operation

3) Method of Graph Plotting: The performance evaluation of the paper-based humidity sensor was carried out by adopting a comparative calibration approach against a commercially available DHT11 humidity sensor. Both sensors were interfaced with an Arduino Uno, which served as the data acquisition unit [46], [47]. The paper sensor was incorporated into a voltage divider circuit, and its resistance values were derived from the measured output voltage. Because the raw output of the paper sensor was found to be prone to fluctuations, each data point was obtained as the average of ten consecutive readings in order to minimize noise and enhance signal stability [46].

For calibration, the averaged resistance values from the paper sensor were matched with the corresponding relative humidity values reported by the DHT11 sensor. The humidity data from the DHT11 were directly plotted as the reference curve, while the paper sensor readings were first processed and then plotted against the same humidity scale. To represent the relationship more clearly, a best-fit curve was applied to both datasets, thereby reducing random variations and highlighting the functional trend of the sensor response [47]. The approach is summarized in Table III.

G. Tank to Mask Miniaturisation

Parallel to the CPAP pump development, the tank based humidity cycling environment has been successfully miniaturised into a wearable nebuliser mask platform integrated with the developed sensing architecture [49], [50]. The optimised humidity sensor has been embedded inside the mask at a location where exhaled airflow produces strong and measurable humidity fluctuations during the respiratory cycle [51]. By recreating the functional characteristics of the controlled humidity chamber

TABLE III: Data processing and plotting approach

Dataset	Processing method	Representation in graph
DHT11 (Reference)	Direct values recorded	Plotted as baseline humidity curve
Paper Sensor	Averaged over 10 samples per point	Plotted against DHT11 humidity with best-fit curve

within the confined mask volume, the system enables real time detection of breathing rate and breathing intensity under practical operating conditions [52].

During the transition from the laboratory humidity chamber to the compact wearable mask environment, the original sensor geometry was redesigned and miniaturised to improve compatibility with the confined respiratory interface.

In order to fit the system inside a nebulizer mask, we had to decrease the overall size of the sensor. However, if we try to squeeze more teeth into this smaller footprint to boost performance, the sensitivity shoots up by too much. While that might seem ideal for detecting incredibly minute humidity changes, it actually makes the device very hard to calibrate.

The root of the issue is architectural. Packing more teeth together leaves less space on the substrate, which accelerates salt deposition during moisture cycling. This buildup triggers severe degradation problems, eating away at the sensor's long-term stability. As the material breaks down, the overall resistance increases and makes precise calibration an uphill battle. We found that our current composition works best because it strikes the perfect balance. It gives us enough sensitivity to track respiration within the compact mask environment without risking early decay or dealing with a volatile baseline.

The modified sensor design adopts a compact serpentine interdigitated graphite structure fabricated on the paper substrate, allowing efficient utilization of the reduced sensing area while maintaining sufficient conductive path length and humidity interaction surface. The redesigned geometry additionally improves flexibility, compact integration, and airflow exposure inside the wearable mask assembly. The dimensional layout, fabrication process, and final fabricated miniaturised sensor structure are illustrated in Fig. 10 and Fig. 11.

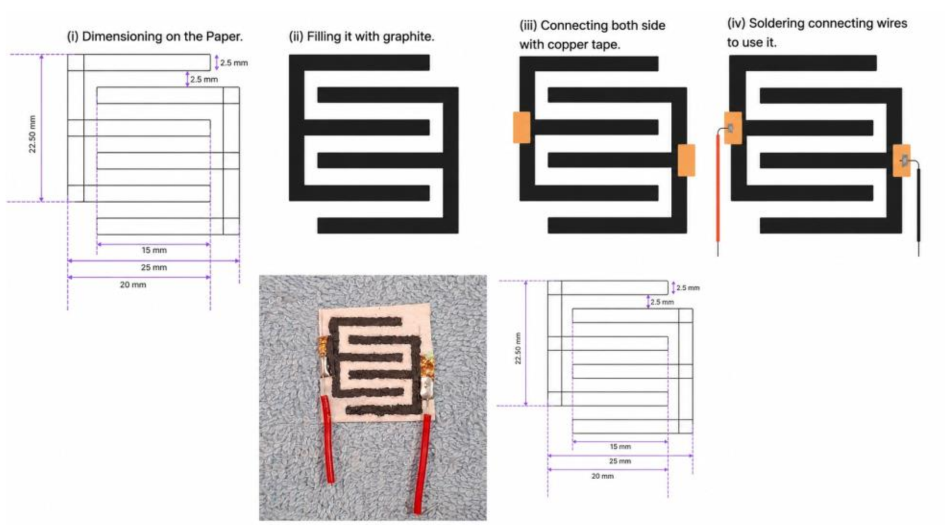
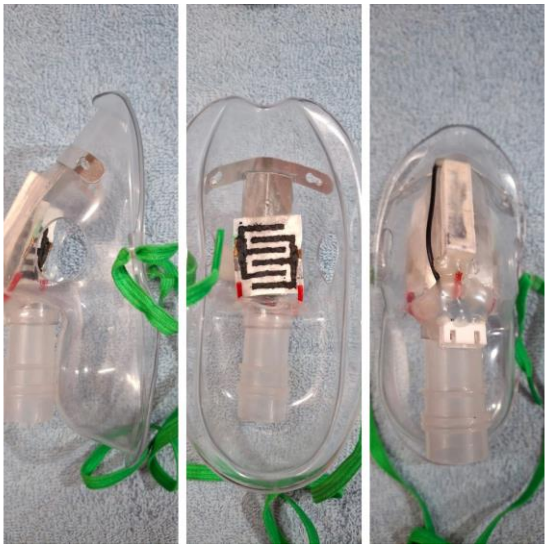


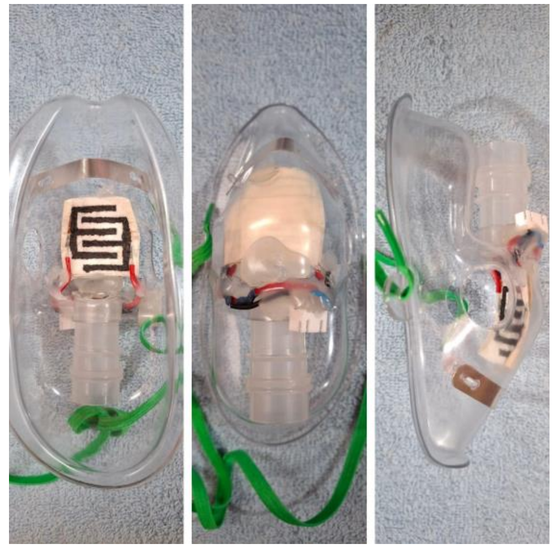
Fig. 10: Miniaturised serpentine humidity sensor design and fabrication procedure for wearable CPAP integration

To improve repeatability and stabilise sensor response during continuous respiratory monitoring, integrated thermal regulation elements have been incorporated into the sensing unit. Two separate thermal stabilization configurations were experimentally implemented. In the first configuration, a compact power resistor-based heating arrangement was used to provide localized heating near the sensing surface. In the second configuration, a serpentine-patterned Nichrome wire heating structure was fabricated and integrated alongside the humidity sensor to generate distributed and uniform thermal control across the sensing region. These heating arrangements assist in reducing moisture accumulation, accelerating sensor recovery, and improving repeatability during repeated inhalation and exhalation cycles.

The complete sensing module, therefore, consists of the humidity sensor integrated with the thermal stabilisation system inside the wearable mask structure. This integrated architecture allows stable capture of breathing-induced humidity transients while minimizing response drift caused by prolonged moisture exposure. The mask simultaneously functions as the sensing chamber and the airflow interface for the CPAP pump, thereby forming a closed-loop respiratory monitoring and pressure regulation system where breathing-driven humidity variations directly influence airflow control and pressure modulation [49], [53]. This miniaturised implementation marks the successful transition from laboratory-scale humidity chamber testing to a wearable respiratory sensing and support platform suitable for future clinical refinement and evaluation.



(a) Power Resistor Configuration



(b) Nichrome Wire Configuration

Fig. 11: Thermal stabilization configurations integrated with the miniaturized humidity sensing unit

The resulting calibration curves allowed a direct comparison between the performance of the paper-based sensor and the industry-standard DHT11 sensor. This comparative approach ensured that the deviations in sensitivity, response profile, and long-term stability could be clearly visualised, thereby validating the effectiveness and practical applicability of the developed paper sensor.

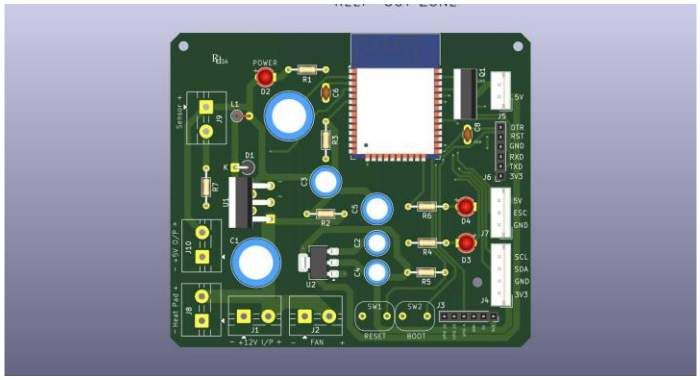
IV. Developed Embedded System and Peripherals

A. Integrated PCB

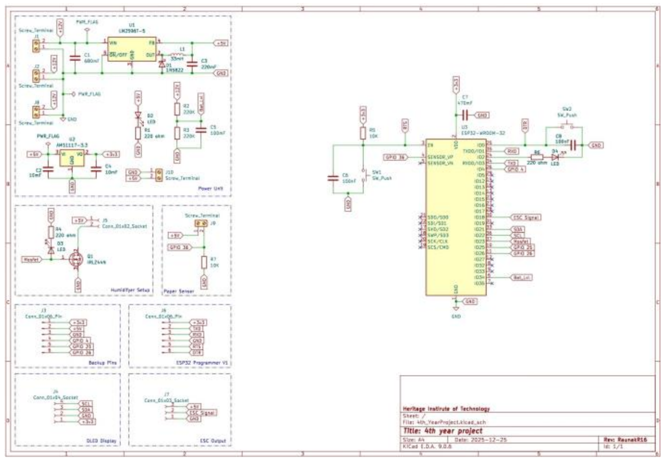
1) System overview: The PCB was designed as a compact embedded platform for the wearable autoregulating CPAP prototype, bringing together sensing, power regulation, actuation, communication, and user feedback in a single integrated hardware system. At the core of the design is the ESP32-WROOM-32 microcontroller, which performs respiratory sensor acquisition, Bluetooth communication, pump regulation, heating control, and OLED display management while supporting both automatic and manual operating modes [54], [55]. The layout was intentionally optimized to reduce wiring complexity, improve electrical reliability, and limit electromagnetic interference that could affect low-level sensor measurements during continuous respiratory monitoring. The schematic and 3d view can be observed in figure 12 [55].

2) Power and control: A higher input voltage is supplied to the airflow subsystem and converted into a regulated 5 V rail through an LM2596T-5 switching converter, which offers better efficiency and lower thermal dissipation than a linear regulator [55], [56]. The 3.3 V supply required by the ESP32 and related low-voltage components is generated using an AMS1117-3.3 regulator, with decoupling capacitors placed close to the supply lines to improve noise rejection and transient stability [55], [57]. This arrangement allows the PCB to support both control circuitry and higher-current actuation elements from a common supply while maintaining stable operation.

3) Sensing, actuation, and interfaces: The paper-based humidity sensor is connected through a voltage-divider circuit and routed to the ESP32 ADC input on GPIO36, allowing breathing-induced resistance changes to be converted into measurable voltage signals [54], [55]. In parallel, the HX710B pressure module provides pressure feedback through a dedicated serial interface, while the OLED display presents local information such as operating mode, pressure, and system status [55], [58]. For airflow generation, the ESP32 produces PWM signals that drive the ESC connected to the BLDC pump, and the firmware applies gradual speed ramping to avoid abrupt airflow transitions and improve patient comfort [55]. The PCB also includes MOSFET-based switching for heater and humidifier functions, which assists with sensor recovery and humidity stabilization during testing [55].



(a) PCB 3D View



(b) PCB Schematic

Fig. 12: PCB 3D and schematic view

B. BLDC PUMP

Instead of relying on expensive imported hardware, the present work focused on developing a fully indigenous autoregulating CPAP platform fabricated entirely within the laboratory environment. In alignment with the “Make in India” initiative, the objective was to create a low-cost, modular, and easily serviceable respiratory support system that prioritizes domestic manufacturing while providing complete component-level accessibility for future modification, maintenance, and scalability [59], [60].

At the core of the developed platform is a custom BLDC-driven turbo impeller pump designed specifically for compact respiratory airflow generation. The developed centrifugal blower consists of an optimised curved blade impeller enclosed within a streamlined volute housing architecture. The pump utilizes a centrally positioned axial inlet for efficient air intake and a tangential outlet nozzle for delivering smooth and pressurised airflow suitable for CPAP operation [61], [62]. The lower section of the housing integrates a compact high-efficiency brushless DC motor mount, thereby enabling stable high-speed operation while maintaining a compact form factor. The entire blower assembly, including the impeller, housing, and airflow channel, was designed, modelled, and fabricated in-house using additive manufacturing techniques and rapid 3D printing methods [63], [64]. This fabrication strategy enabled rapid prototyping, iterative aerodynamic optimization, and a significant reduction in manufacturing cost.

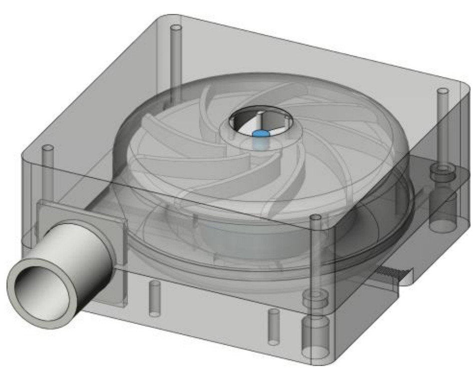
The developed custom blower successfully generated the airflow and pressure ranges required for CPAP assisted respiratory support while remaining sufficiently compact for wearable and portable operation. By eliminating dependence on proprietary imported blower modules, the system addresses one of the most expensive components found in commercial CPAP devices and directly reduces the overall system cost [65]. Furthermore, the indigenous development of the pump architecture breaks dependence on imported respiratory hardware and contributes toward decentralised low-cost healthcare technology development.

The integration of the paper based humidity sensing platform together with the developed BLDC turbo impeller pump forms a closed loop autoregulating respiratory assistance system capable of responding dynamically to patient breathing behaviour in real time. The resulting indigenous platform demonstrates the feasibility of combining sustainable humidity sensing technology

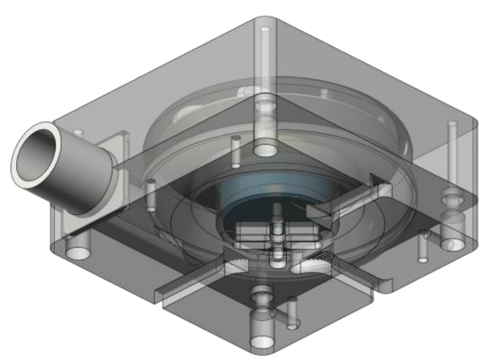
Component	Key specification(s)	Primary use / connection
ESP32-WROOM-32	Dual-core MCU, WiFi, Bluetooth, ADC1 GPIO36, multiple UART, I^2C, and SPI pins	Central processor for sensor acquisition, ADC input on GPIO36, I^2C for HX710B and OLED, UART/Bluetooth telemetry, and PWM for ESC [54], [55]
LM2596T-5	5 V fixed step-down switching regulator, up to 3 A typical, requires external L1, D1, and capacitors	Converts battery input to a stable 5 V rail for peripherals and logic [55], [56]
AMS1117-3.3	3.3 V linear LDO, up to 1 A, with recommended decoupling	Supplies 3.3 V to ESP32 and other low-voltage peripherals [55], [57]
Paper humidity sensor	Variable-resistance hygroscopic paper element integrated in a voltage divider to produce ADC-compatible voltage at GPIO36	Breath-driven humidity sensing front end for respiratory waveform extraction [55]
HX710B pressure module	24-bit pressure transducer module, typical 0–40 kPa range, 3.3–5 V logic interface	High-resolution pressure sensing connected to dedicated serial or I^2C lines [55], [58]
BLDC pump and ESC	ESC accepts PWM pulses of about 1074 μs to 1600 μs, with firmware ramping to avoid abrupt changes	Generates controlled airflow; ESC signal is routed from MCU PWM [55]
IRLZ44N MOSFET	Logic-level N-channel power MOSFET, V_{DS} about 60 V, low $R_{DS(on)}$ at V_{GS}=5 V	Switches heater elements and other high-current loads [55]
D4184 MOSFET	Logic-level MOSFET used in the heating switch stage	Controls resistor or nichrome heating elements for sensor recovery [55]
1N5822 Schottky diode	Low forward-voltage high-current Schottky diode used in switching regulator stage	Freewheeling and rectification in LM2596 buck converter [55]
L1 inductor	About 33 μH inductor used in the buck converter	Energy storage and smoothing for LM2596 output [55]
Decoupling capacitors	100 nF ceramics and bulk electrolytics near regulators and MCU	Suppress ripple and transient noise for stable ADC readings [55]
OLED 0.96 inch	Small graphical display, typically I^2C interface	Local user interface for mode, telemetry, and status [55]
Screw terminals and headers	Mechanical connectors for sensors, pump, OLED, battery, and programmer	Modular connection points that simplify debugging and replacement [55]
Push switches	Mechanical buttons with pull resistors for user input	Mode selection and local commands [55]

TABLE IV: Primary PCB components and functions

with locally manufactured respiratory support hardware to create an accessible, scalable, and self reliant healthcare solution suitable for resource constrained medical environments, The 3d view is presented in figure 13 [66], [67].



(a) BLDC Pump 3D Design – Top View



(b) BLDC Pump 3D Design – Rear View

Fig. 13: BLDC Pump 3D CAD Model – Top and Rear Views

1) Pressure Calibration: From the graph fig. 14 , there is a direct and positive correlation between the PWM control signal input of the BLDC motor and the output pressure generated. When tested at an environmental pressure of 1004.94 hPa, the tuned fan design produces a base pressure of 0.00 hPa (0.00 cmH2O) at a low PWM input of 1074. However, the output pressure rises gradually when the PWM input is increased up to a maximum PWM value of 2000, achieving a maximum output pressure of 18.87403 hPa (19.24615 cmH2O).

Upon analysis of the pressure response curve, it shows a nonlinear relationship featuring varied rates of change in different operational zones. At a low PWM operation, the pressure response exhibits an acute pressure increment from PWM 1166 to 1259. In turn, the increasing pressure response stabilizes with a gradual rise in pressure at intermediate operation ranging from PWM 1351 to 1722. Finally, the pressure generation accelerates again at high operational rates leading to maximum pressure.

The above behavior affects how the fan can be incorporated into CPAP applications, where the pressure delivery should be between 4 and 30 cmH2O. After tuning the system, it was found that the system was able to satisfy a large range of pressures, which is 0.00 cmH2O to 19.24615 cmH2O, within the therapeutic delivery range required. The data table is shown in table V and the graph is shown in fig. 14

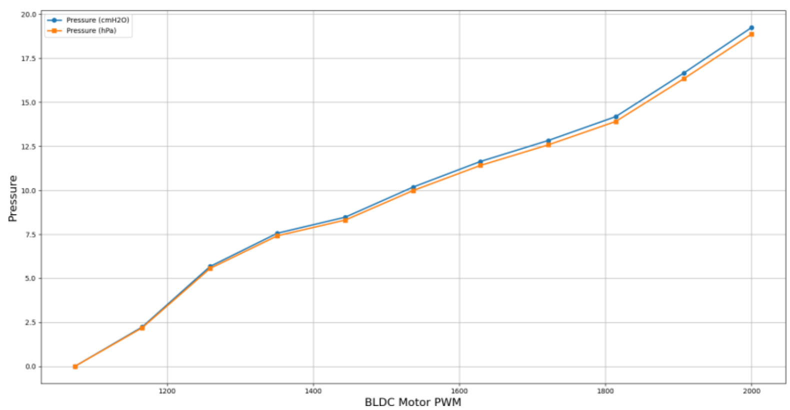


Fig. 14: Graph of PWM vs Pressure performance

TABLE V: BLDC Motor PWM and Corresponding Air Pressure Output

Fan Speed Level	BLDC Motor PWM	Air Pressure (hPa)	$cmH_2O = hPa \times 1.019716$
0	1074	0.00	0.00
1	1166	2.18684	2.2299521492
2	1259	5.56207	5.6717282592
3	1351	7.41330	7.5594655233
4	1444	8.30771	8.4715093606
5	1537	9.98726	10.1841696528
6	1629	11.41118	11.6361643034
7	1722	12.58217	12.8302356974
8	1814	13.907001	14.1811967992
9	1907	16.33036	16.652397982
10	2000	18.87403	19.2461475735

C. Control Algorithm for Adaptive CPAP Regulation

The operational sequence of the developed autoregulating CPAP platform begins when the patient breathes normally through the respiratory mask interface. During exhalation, humid air released from the nostrils directly interacts with the active humidity sensing element integrated inside the mask chamber. The sensing element consists of a graphite coated paper based humidity sensor whose electrical resistance changes predictably in response to moisture exposure [68], [69]. To convert these resistance variations into measurable electrical signals, the sensor was integrated into a voltage divider configuration that produces corresponding analog voltage fluctuations. These analog signals are continuously acquired by the internal Analog-to-Digital Converter (ADC) of the ESP32 microcontroller and converted into digital values for real time processing and analysis [70].

1) Sensor Conditioning and Thermal Recovery Loop: To maintain sensing stability and prevent long term signal degradation during continuous respiratory monitoring, a dedicated sensor conditioning and thermal recovery loop was implemented. Continuous exposure to exhaled moisture can gradually saturate the sensing substrate because of water accumulation within the porous cellulose structure. This saturation effect leads to delayed recovery, baseline drift, and loss of sensing accuracy

[71]. To overcome this limitation, the developed control algorithm continuously monitors the sensor response characteristics and evaluates the recovery behaviour of the substrate.

When saturation or excessive moisture retention is detected, the control system automatically activates an integrated thermal regulation circuit positioned adjacent to the sensing region. The thermal recovery mechanism utilizes localized heating elements to dry the sensor surface and accelerate moisture desorption from the cellulose fibre network. Once the sensor returns to its baseline operating condition, the heating cycle is terminated automatically and the sensing element becomes ready for the next respiratory cycle. This closed loop thermal recovery architecture significantly improves repeatability, minimizes hysteresis, and enhances long term operational stability during continuous respiratory monitoring [72], [73].

2) Adaptive CPAP Airflow Regulation Loop: Operating simultaneously with the thermal conditioning cycle is the adaptive airflow regulation loop responsible for autoregulating CPAP pressure delivery according to the patient's real time breathing behaviour. The ESP32 microcontroller continuously analyses incoming humidity response signals and identifies sudden increases in humidity generated during exhalation. Each detected humidity spike is interpreted as a distinct respiratory event and converted into pulse like digital breathing signals for respiratory pattern analysis [74].

The controller continuously counts these respiratory events over time to determine the patient's real time breathing rate in breaths per minute. The calculated respiration rate is then compared against predefined physiological breathing thresholds corresponding to normal respiratory conditions, typically ranging between 12 and 20 breaths per minute [75]. Based on this comparison, the controller dynamically adjusts the operating speed of a brushless DC motor connected to the developed turbo impeller pump assembly.

Under stable breathing conditions, the system reduces CPAP delivery pressure by lowering the BLDC motor speed, thereby maintaining comfortable airflow conditions while minimizing unnecessary pressure loading on the patient. Conversely, when abnormal breathing patterns or respiratory instability are detected, the controller immediately increases airflow support by accelerating the BLDC motor speed. The resulting increase in CPAP pressure provides enhanced pneumatic assistance to maintain airway patency and stabilize respiration [66], [76].

The generated pressurized airflow is delivered directly to the patient through the respiratory mask interface. Simultaneously, the increased airflow assists in clearing excess moisture accumulation from the sensing chamber and indirectly influences subsequent humidity measurements. As a result, the humidity sensing system, thermal recovery mechanism, and adaptive airflow controller collectively form a fully integrated smart closed loop respiratory support architecture capable of dynamically responding to patient breathing behaviour in real time [77].

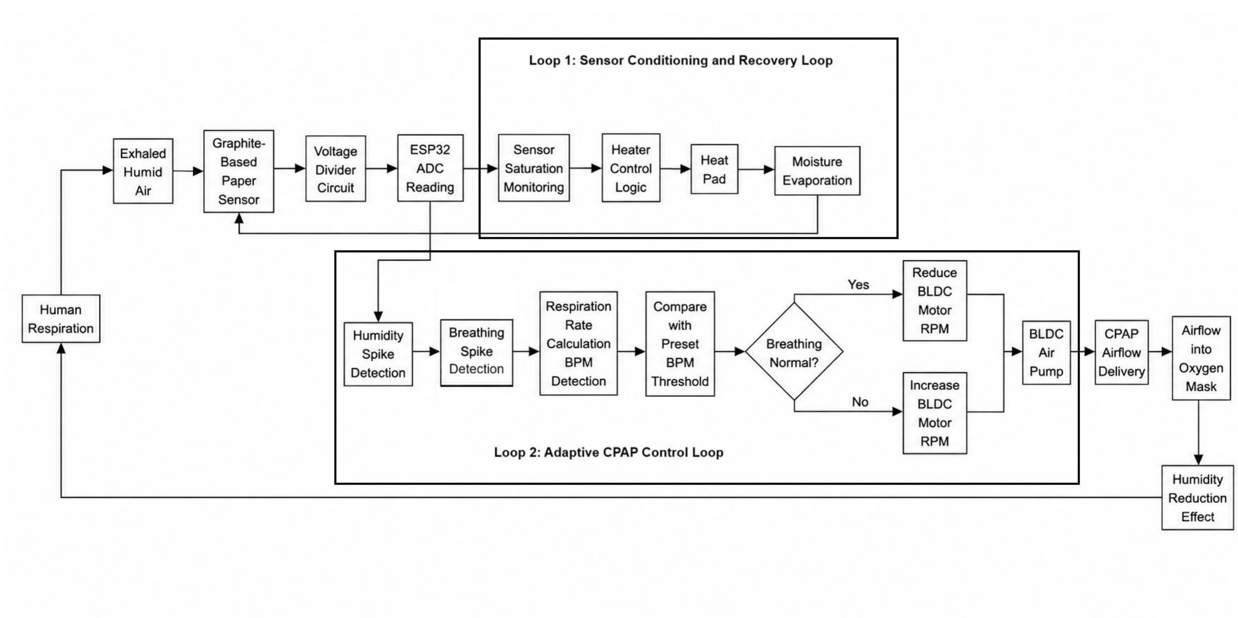


Fig. 15: Control Architecture of the entire system

D. Wireless Connectivity, Edge Visualization, and Mobile Interface Design

Wireless communication capability was integrated into the developed autoregulating CPAP platform using the ESP32 microcontroller. The embedded firmware establishes a Bluetooth classic communication under the broadcast device identifier "BreathEZ," thereby creating a wireless bridge between the embedded control hardware and external monitoring interfaces [78],

[79]. Through this communication channel, the operational state of the system can be wirelessly controlled, allowing seamless switching between automatic closed-loop regulation mode and manual operation mode. In addition, the wireless architecture continuously transmits real-time sensor data streams, enabling remote recalibration commands, pump speed adjustment, and live respiratory monitoring during experimental evaluation and diagnostic operation.

1) Edge Visualization and Standalone Subsystems: To improve standalone operation and reduce dependence on external computing hardware during localized testing, an Organic Light Emitting Diode (OLED) display interface was integrated directly onto the device. The OLED subsystem provides immediate visual feedback regarding the operational state of the device and continuously displays critical system parameters, including the active operating mode, pump speed indices, real-time air pressure values, substrate temperature, and humidifier status [80], [81]. The integration of this dedicated local display significantly improves system portability, testing ergonomics, and independent operation capability during wearable respiratory monitoring applications.

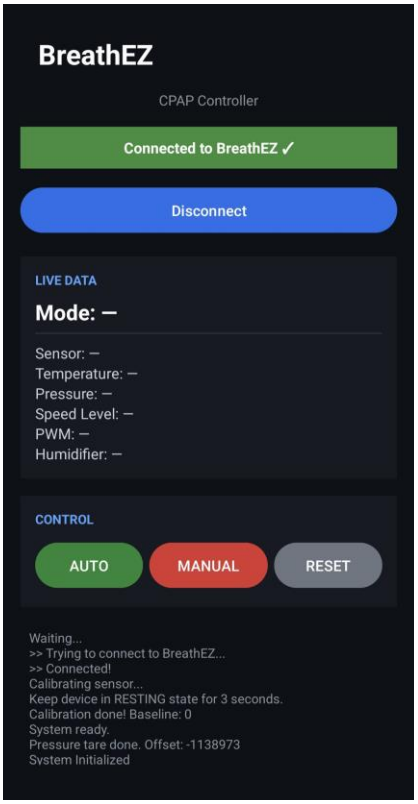


Fig. 16: Application Dashboard

2) Dedicated Mobile Application and Remote Interface Architecture: To extend system accessibility beyond the physical device interface, a dedicated Android-based mobile application was developed using Android Studio. The application communicates directly with the ESP32 controller through the established Bluetooth classic communication protocol and functions as a portable wireless control and monitoring dashboard [82], [83]. The graphical user interface was designed to simplify operational control by enabling rapid switching between manual and automatic operating modes through a smartphone interface shown in figure 16.

In addition to operational mode control, the application provides real time visualization of critical system variables including instantaneous pump speed, humidity sensor response behaviour, dynamic pneumatic pressure variation, and auxiliary humidifier status. By combining wireless telemetry, embedded visualization, and smartphone based monitoring, the developed platform achieves improved portability, operational flexibility, and decentralized usability suitable for practical wearable respiratory support applications [84], [85].

V. RESULTS AND DISCUSSION

The humidity sensing performance of the fabricated substrates was systematically analysed under repeated humidity cycling conditions in order to identify the most suitable sensing platform for respiratory monitoring and CPAP integration. The comparison was divided into two stages. Initially, a quantitative comparison was performed between the handmade paper sensor and the Whatman Grade 1 filter paper sensor using experimentally calculated electrical response parameters. Following the selection of the optimized paper substrate, the polypropylene cloth sensor was comparatively evaluated through microscopic surface analysis and humidity response behaviour to establish the influence of substrate morphology on sensing performance.The obsereved graphs are presented in figures: 19, 20 and 21.

A. Humidity Response of the sensors

The humidity response of the two paper substrates follows the behaviour observed in Fig. 19. Handmade paper shows a sharp and rapid drop in resistance as humidity rises from moderate to high levels because its porous structure absorbs moisture quickly. Normal paper shows a slower change during the early stage of humidity increase and only displays a rapid drop when humidity approaches saturation. The difference in response rate reflects the underlying fibre structure of the two papers [24], [86], [87].

The polypropylene cloth sensor, however, exhibits a slower and more gradual response to rising humidity because polypropylene is naturally hydrophobic. Moisture interaction occurs mainly at the surface rather than through bulk absorption, resulting in smoother but less pronounced resistance variation compared to the paper substrates [24], [87], [88].

1) Electrode Trace Degradation Rate: The degradation of graphite traces differs noticeably between the substrates. Handmade paper shows faster deterioration during repeated humidity cycles, which can be observed in the graphs of Fig. 19. This occurs because its irregular fibre network retains more moisture, progressively disrupting the conductive pathways. Normal paper maintains the graphite layer for longer durations because its surface is smoother and absorbs moisture more uniformly, delaying resistance drift [24], [86], [88].

The polypropylene cloth substrate demonstrates improved structural durability because the fibres do not absorb moisture or undergo swelling. However, the hydrophobic surface weakens graphite adhesion and may produce localized instability or patchy conductivity after repeated flexing cycles [24], [87].

2) Operational Cycles and Stability: Both paper based sensors operate reliably for approximately one complete humidity cycle without thermal assistance. When the heat pad is applied, the operational life extends to nearly three cycles because the heating reduces moisture retention and limits paper deformation, allowing the resistance to recover closer to its original value during drying, as observed in Fig. 20 [24], [86].

The polypropylene cloth sensor demonstrates longer operational life because the substrate does not swell or deform during humidity exposure. Although physically stable, the electrical response remains less consistent because the overall resistance variation produced by surface level moisture interaction is comparatively small [87], [88].

3) Sensitivity to Small Humidity Changes: Handmade paper exhibits higher sensitivity to small humidity variations, particularly at lower humidity levels where resistance changes occur rapidly with minor moisture fluctuations. Normal paper is comparatively less responsive in this region and requires larger humidity changes before significant resistance variation becomes observable. This difference originates from the comparatively lower porosity and denser fibre distribution of the normal paper substrate [24], [87], [88].

The cloth sensor shows the lowest sensitivity among the investigated substrates because moisture interaction is restricted primarily to surface adsorption. Consequently, the resulting resistance variation is smaller and less suitable for detecting fine respiratory humidity fluctuations required in breath monitoring applications [87], [88].

4) Resistance Increase Mechanism and Overall Behaviour: For both paper substrates, the increase in resistance after repeated humidity cycling is associated with fibre swelling and disruption of the graphite conductive pathways. Handmade paper experiences stronger swelling effects because of its highly porous structure, whereas normal paper undergoes similar degradation mechanisms to a lesser extent due to its comparatively compact and uniform fibre network [24], [86], [88].

5) Microscopic Surface Analysis of Sensor Substrates: Microscopic analysis of the sensor surfaces was performed to establish a direct link between substrate morphology and the experimentally observed humidity sensing behaviour. As shown in Fig. 17, the paper based sensor surface consists of a dense and highly irregular cellulose fibre network with pronounced surface roughness and high porosity. The fibres are short, randomly oriented, and tightly packed, forming interconnected pores and capillary channels that enable rapid moisture absorption and diffusion throughout the substrate. Graphite traces appear mechanically anchored within the fibrous matrix due to fibre entanglement and surface asperities, resulting in initially continuous conductive pathways. However, repeated humidity cycling leads to fibre swelling, local structural distortion, and micro crack formation within the graphite layer, progressively disrupting electrical continuity. This microstructural behaviour directly explains the sharp resistance variation, high sensitivity at low relative humidity, and accelerated degradation observed in paper based sensors during prolonged operation [89]–[92].

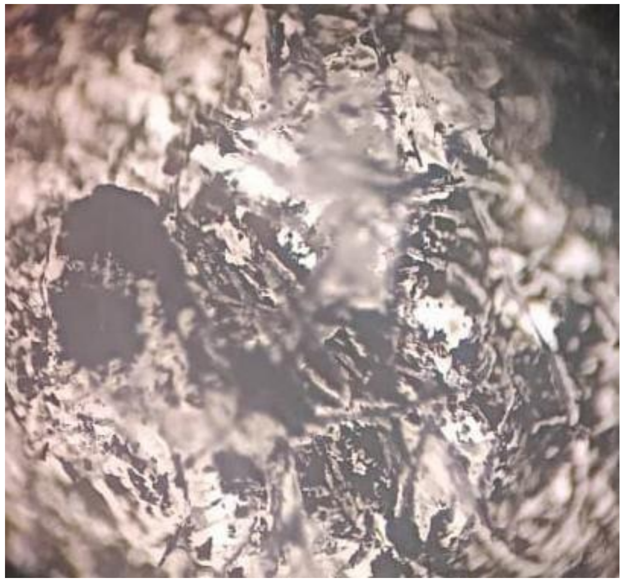


Fig. 17: Microscopic image of the paper based sensor surface showing dense cellulose fibre networks, high porosity, and irregular morphology that promote moisture absorption and graphite anchoring.

In contrast, the polypropylene cloth sensor surface shown in Fig. 18 exhibits a loosely entangled network of long, smooth synthetic fibres with relatively uniform diameters and large inter fibre voids. The fibres lack intrinsic porosity and capillary structures, confirming the hydrophobic nature of polypropylene and its resistance to bulk moisture uptake [93]. Graphite deposition is primarily confined to the outer surfaces of individual fibres and fibre junctions, leading to partially discontinuous conductive pathways. As moisture interaction is limited to surface adsorption and minor contact resistance modulation, the resulting resistance change with humidity is gradual and of low amplitude. This microstructure accounts for the superior mechanical stability and extended operational life of the cloth sensor, while also explaining its reduced sensitivity and occasional electrical irregularity under repeated humidity cycling [94], [95].

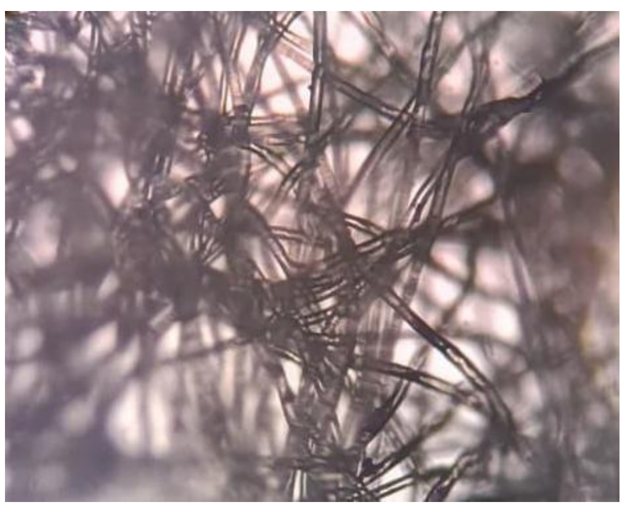
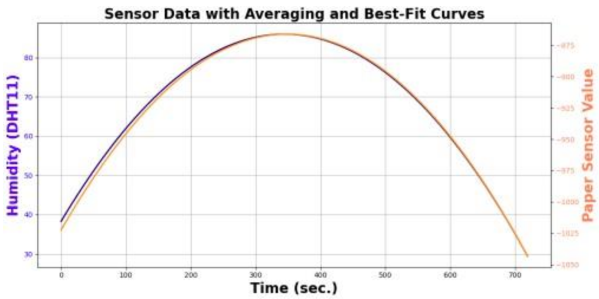
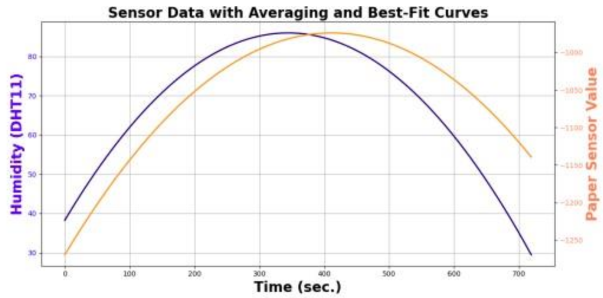


Fig. 18: Microscopic image of the polypropylene cloth sensor surface illustrating smooth synthetic fibres, large inter fibre gaps, and surface confined graphite deposition.

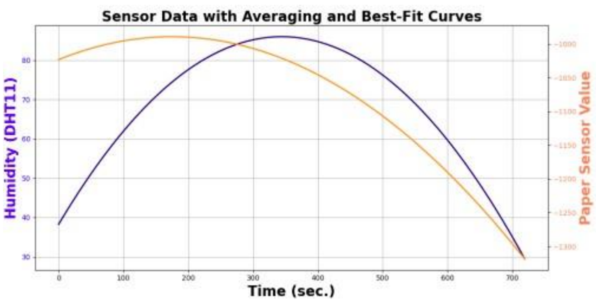
Overall, the microscopic surface characterization establishes a clear structure performance relationship for the investigated humidity sensors. The porous cellulose architecture of paper substrates enables strong moisture interaction through bulk absorption and fibre swelling, resulting in high sensitivity but limited long term stability. Conversely, the smooth and non porous synthetic fibre structure of polypropylene cloth preserves mechanical integrity and durability, but inherently restricts moisture interaction and sensing resolution. The microstructural features observed in Fig. 17 and Fig. 18 therefore validate the macroscopic electrical measurements and clearly demonstrate the fundamental trade off between sensitivity and stability that governs substrate selection for resistive humidity sensing applications [96].



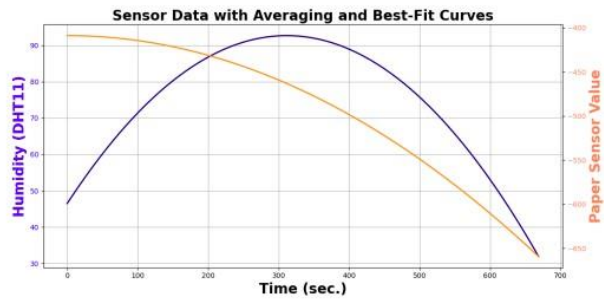
Graph 01: Grade 1 Filter Paper at first use.



Graph 03: Hand made Filter Paper at first use.

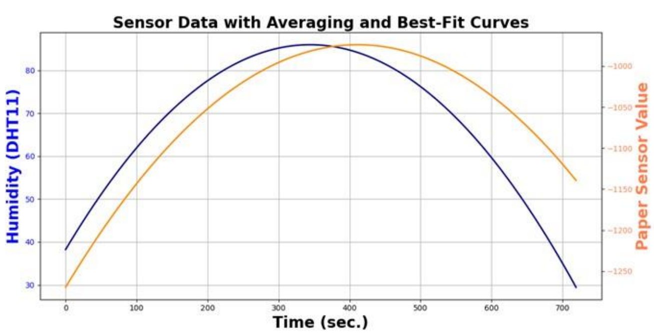


Graph 02: Grade 1 Filter Paper after several use.

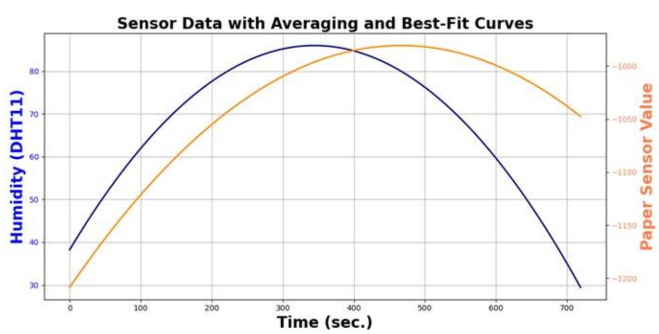


Graph 04: Hand made filter paper after several use.

Fig. 19: Various graphs showing performance analysis of paper based sensors vs DHT sensor (With Heat Pad)



Graph 05: Grade 1 filter paper at first use.



Graph 05: Hand made filter paper at first use.

Fig. 20: Graphs showing performance analysis of paper-based sensors vs DHT sensor (Without Heat Pad)

Overall, the polypropylene cloth sensor demonstrates superior mechanical durability and resistance to structural degradation, while the paper based substrates exhibit stronger humidity sensitivity. This establishes a clear trade off between sensing performance and long term stability. The paper substrates provide stronger humidity responsiveness suitable for respiratory sensing applications, whereas the cloth substrate offers improved durability but reduced sensing resolution.

B. Mathematical Calculations and Analysis

The experimental characterization of the paper-based humidity sensors involves the systematic measurement of multiple performance metrics to evaluate response dynamics, cyclic stability, and thermal modulation effects [97]. The core parameters

TABLE VI: Comparison of Sensor Characteristics on Hand-Made and Normal Paper Substrates

Parameter	Hand Made Paper	Normal Paper	Remarks
Thickness (mm)	0.12 – 0.5	0.17 – 1.00	Hand made paper is thinner and more porous
Surface Morphology	Irregular fibre structure, high porosity	Smooth and uniform fibre distribution	Influences moisture absorption and electrode adhesion
Initial Sensor Resistance (R_o)	30 kΩ to 70 kΩ	30 kΩ to 70 kΩ	Comparable initial resistance across both substrates
Humidity Response	Sharp and rapid increase between 50%-100% RH	Slower increase between 50%-80%, then rapid near 100% RH	Hand made paper more sensitive to high RH changes
Electrode Trace Degradation Rate	Faster degradation observed	Slower degradation observed	Due to moisture absorption and mechanical stress
Operational Cycles Without Heat Pad	Functional for 1 full cycle	Functional for 1 full cycle	Paper deformation causes resistance to exceed 1 MΩ
Operational Cycles With Heat Pad	Approximately 3 cycles	Approximately 3 cycles	Heat pad reduces physical deformation, extending life
Sensitivity to Small Humidity Changes	High sensitivity at low RH	Limited sensitivity at low RH	Important for applications like breath monitoring
Resistance Increase Mechanism	Attributed to swelling and lattice changes	Same	Structural and molecular changes disrupt conductivity

TABLE VII: Comparison of Best Performing Paper Sensor and Polypropylene Cloth Sensor

Parameter	Best Paper Sensor	Cloth Sensor	Remarks
Substrate Behaviour	Absorbs moisture rapidly	Hydrophobic and moisture resistant	Influences sensing mechanism
Humidity Response	Sharp and noticeable change with RH	Slow and shallow change with RH	Paper is more sensitive to high RH
Sensitivity to Small Changes	High sensitivity especially at low RH	Low sensitivity	Moisture penetration differs
Mechanical Stability	Can swell and deform over cycles	Highly flexible, no swelling	Cloth retains shape better
Electrode Stability	Graphite degrades with moisture	Adhesion may be weak but substrate stable	Long-term behaviour depends on surface bonding
Operational Cycles With Heat Pad	Approximately 3 cycles	More than 3 cycles	Cloth structure does not deform
Consistency of Readings	Good initially but drifts after swelling	Physically stable but electrical response irregular	Stability vs sensitivity trade-off
Overall Advantage	Strong humidity response	High durability	Depends on required application
Overall Limitation	Limited longevity due to deformation	Low response amplitude	Cloth unsuitable for fine detection

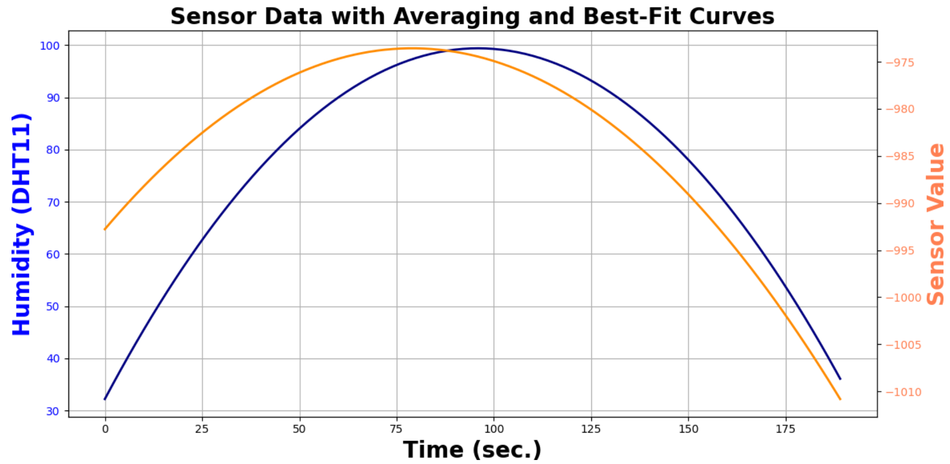


Fig. 21: Graph of DHT vs Cloth sensor perfromance

quantified during the trials include the absorption sensitivity denoted as S_{ADS} and measured in ADC/%RH, the desorption sensitivity denoted as S_{DES} and measured in ADC/%RH, and the recovery slope ratio designated as R expressed as a percentage.

The sensitivity parameters quantify the sensor's signal response per unit change in relative humidity during the moisture uptake and release phases, expressed as:

$$S_{\mathrm{ADS}} = \frac{\Delta \mathrm{ADC}_{\mathrm{absorption}}}{\Delta \%\mathrm{RH}} \tag{3}$$

$$S_{\mathrm{DES}} = \frac{\Delta \mathrm{ADC}_{\mathrm{desorption}}}{\Delta \%\mathrm{RH}} \tag{4}$$

Additionally, the geometric behavior of the signal is captured through the increase in slope represented by $A(n)$ and the decrease in slope represented by $B(n)$, where n signifies the respective experimental state or configuration index. The recovery slope ratio R evaluates the relationship between these transient response geometry features:

$$R = \left(\frac{B(n)}{A(n)}\right) \times 100\% \tag{5}$$

These parameters collectively determine how efficiently the sensor tracks the rapid moisture fluctuations associated with human respiration patterns [98].

The first phase of data collection focuses on the Grade-1 filter paper sensor under varying conditions of use and thermal assistance. For the Grade-1 filter paper after its first use with an integrated heat pad, the absorption sensitivity is recorded at -3.4902, the desorption sensitivity is -3.9375, the recovery slope ratio is 106.1798%, with an increasing slope $A(1)$ of 55 and a decreasing slope $B(1)$ of 356. Following extended cycling, the Grade-1 filter paper after 10 uses with the heat pad demonstrates an absorption sensitivity of -1.0785, a desorption sensitivity of -6.0339, a recovery slope ratio of 647.2728%, an increasing slope $A(2)$ of 178, and a decreasing slope $B(2)$ of 189. In the absence of thermal assistance, the Grade-1 filter paper after its first use without a heat pad exhibits an absorption sensitivity of -3.490, a desorption sensitivity of -3.9375, a recovery slope ratio of 106.1798%, an increasing slope $A(3)$ of 178, and a decreasing slope $B(3)$ of 184.

The second phase evaluates the performance of the handmade paper substrate under a parallel set of experimental conditions to contrast its structural properties with the standardized filter paper. The handmade paper after its first use with a heat pad yields an absorption sensitivity of -6.3462, a desorption sensitivity of -2.5667, a recovery slope ratio of 46.667%, alongside an increasing slope $A(4)$ of 330 and a decreasing slope $B(4)$ of 1034. After undergoing 10 operational cycles with the heat pad, the handmade paper experiences total degradation in its absorption capacity, resulting in an absorption sensitivity of 0, a desorption sensitivity of -4.1269, a highly elevated recovery slope ratio of 26000%, an increasing slope $A(5)$ of 0, and a decreasing slope $B(5)$ of 260. Furthermore, when tested completely without a heat pad, the fresh handmade paper displays an absorption sensitivity of -5.0677, a desorption sensitivity of -2.5970, a recovery slope ratio of 58.193%, an increasing slope $A(6)$ of 299, and a decreasing slope $B(6)$ of 174.

To quantify the long-term stability and structural decay of the sensing layers, the percentage baseline drift and the percentage retention of the directional slopes are calculated using specific mathematical formulations [99]. The percentage drift represents the shift in baseline operational parameters over multiple cycles, reflecting the material robustness against environmental hysteresis, and is modeled as:

$$\text{Percentage Drift} = \left(\frac{X_{\mathrm{final}} - X_{\mathrm{initial}}}{X_{\mathrm{initial}}}\right) \times 100\% \tag{6}$$

where X denotes the specific baseline operational parameter being evaluated (such as baseline capacitance, resistance, or raw ADC output voltage).

The percentage retention for the increasing and decreasing phases of the response slopes isolates the degradation or enhancement of response metrics between different operational states or environmental treatments, formally modeled by the following equations:

$$\text{Percentage Retention (Increasing Slope)} = \left(\frac{A_{\mathrm{final}}}{A_{\mathrm{initial}}}\right) \times 100\% \tag{7}$$

$$\text{Percentage Retention (Decreasing Slope)} = \left(\frac{B_{\mathrm{final}}}{B_{\mathrm{initial}}}\right) \times 100\% \tag{8}$$

These formulas isolate the precise impacts of mechanical wear, moisture retention, and thermal rejuvenation on the sensor substrates.

The evaluation of these formulas reveals stark contrasts in the operational degradation between the two substrate materials under cyclic testing and thermal modulation. The percentage drift after 10 uses for the Grade-1 filter paper with a heat pad is calculated to be -0.0976%, demonstrating exceptional baseline stability, whereas the handmade paper with a heat pad exhibits a massive percentage drift of 68.3076% over the same period. For the Grade-1 filter paper with a heat pad, the percentage retention after 10 uses while the slope is increasing equals 323.63%, and while the slope is decreasing equals 53.089%. Conversely, the handmade paper with a heat pad retains 0% of its increasing slope capability after 10 uses and only 25.096%

of its decreasing slope performance. When examining the direct influence of the heat pad after a single use, the Grade-1 filter paper shows an increasing slope retention of 323.63% and a decreasing slope retention of 51.685% for the unheated versus heated state, while the handmade paper exhibits an increasing slope retention of 90.606% and a decreasing slope retention of 16.827%.

Based on a comprehensive review of the empirical evidence, the Grade-1 filter paper operating with an integrated heat pad emerges as the clear best-performing sensor configuration for breathing pattern detection. The handmade paper substrate demonstrates an unacceptable level of instability, highlighted by its high percentage drift of 68.3076% and the complete failure of its absorption mechanisms after 10 uses, where the absorption sensitivity drops to zero. This catastrophic loss of functionality causes an unmanageable spike in the recovery slope ratio to 26000%, indicating that handmade paper cannot withstand repeated humidity cycles. In sharp contrast, the Grade-1 filter paper maintains a near-zero baseline drift of −0.0976% over 10 uses, indicating superb structural resilience and minimal hysteresis. Although its slope characteristics shift due to continuous exposure, the presence of the heat pad successfully prevents moisture accumulation and ensures that the desorption sensitivity remains high at −6.0339 , thereby confirming that the Grade-1 filter paper with thermal activation is the optimal choice for a reliable, auto-calibrating CPAP diagnostic system [100].The comparison table with all the parameters are given in table VIII

TABLE VIII: Comparative Performance Analysis of Paper Based Humidity Sensors

Sl. No.	**Type of Sensor**	**Absorption Sensitivity** S_{ADS} **(ADC/%RH)**	**Desorption Sensitivity** S_{DES} **(ADC/%RH)**	**Recovery Slope Ratio** $R(\%)$	**Increase in Slope** $A(n)$	**Decrease in Slope** $B(n)$
1.	Grade 1 Filter Paper After 1 Use (With Heat Pad)	-3.4902	-3.9375	106.1798	55	356
2.	Grade 1 Filter Paper After 10 Use (With Heat Pad)	-1.0785	-6.0339	647.2728	178	189
3.	Grade 1 Filter Paper After 1 Use (Without Heat Pad)	-3.490	-3.9375	106.1798	178	184
4.	Hand Made Paper After 1 Use (With Heat Pad)	-6.3462	-2.5667	46.667	330	1034
5.	Hand Made Paper After 10 Use (With Heat Pad)	0	-4.1269	26000	0	260
6.	Hand Made Paper (Without Heat Pad)	-5.0677	-2.5970	58.193	299	174

C. Breathing rate

1) Real Time Respiratory Waveform Analysis: The real-time respiratory waveform obtained from a 74-year-old female patient, as shown in Fig. **??**, demonstrates the practical capability of the developed humidity sensing platform to monitor breathing activity under clinical conditions. The recorded waveform represents a continuous time series response where the vertical axis corresponds to the raw sensor output in ADC counts and the horizontal axis represents time progression. Distinct repetitive peaks and troughs corresponding to complete respiratory cycles are clearly observed throughout the monitoring period. During exhalation, the introduction of humid air into the sensing chamber alters the resistance of the graphite coated sensing layer, resulting in a sudden variation in the measured signal. This response is followed by a rapid recovery phase corresponding to moisture desorption from the substrate surface. A spike detection algorithm successfully identifies the maxima of each respiratory pulse, thereby enabling reliable differentiation between true breathing events and background signal noise [74], [101].

Analysis of the temporal distribution and density of the detected peaks indicates a significantly elevated respiratory rate ranging between approximately 22 and 26 breaths per minute. Such an increased breathing frequency is clinically relevant in elderly individuals experiencing respiratory distress or asthma exacerbation, where insufficient alveolar ventilation and restricted airflow force the body to compensate through rapid breathing cycles [75], [102]. The compressed waveform distribution observed in the recorded signal therefore reflects the physiological stress associated with obstructed respiratory airflow and supports the clinical requirement for immediate nebulization assisted respiratory therapy.

Detailed examination of individual waveform profiles further reveals a distinct asymmetry between the adsorption and desorption phases of the humidity response cycle. The adsorption phase associated with inhaled moisture exposure appears sharp and abrupt, whereas the desorption phase exhibits a comparatively elongated recovery tail. This waveform behaviour closely correlates with the pulmonary pathophysiology associated with asthma and obstructive airway disorders, where expiration becomes prolonged because of bronchial constriction, airway inflammation, and increased expiratory resistance [103]. Consequently, the elongated recovery portion of the waveform provides indirect physiological evidence of increased exhalation difficulty in the monitored patient.

An important observation from the overall waveform trend is the remarkable long term stability of the sensing response despite continuous exposure to highly humid conditions generated during nebulizer operation. No evidence of signal saturation, moisture flooding, or baseline collapse was observed during prolonged monitoring. Instead, individual respiratory peaks remained distinct, separated, and clearly identifiable throughout the experiment. This result experimentally validates the effectiveness of the integrated thermal stabilization system in regulating the local microenvironment surrounding the sensing substrate. The combined heater-assisted configuration successfully prevented droplet accumulation within the paper matrix and preserved stable sensor operation under continuous respiratory humidity exposure [72], [73].

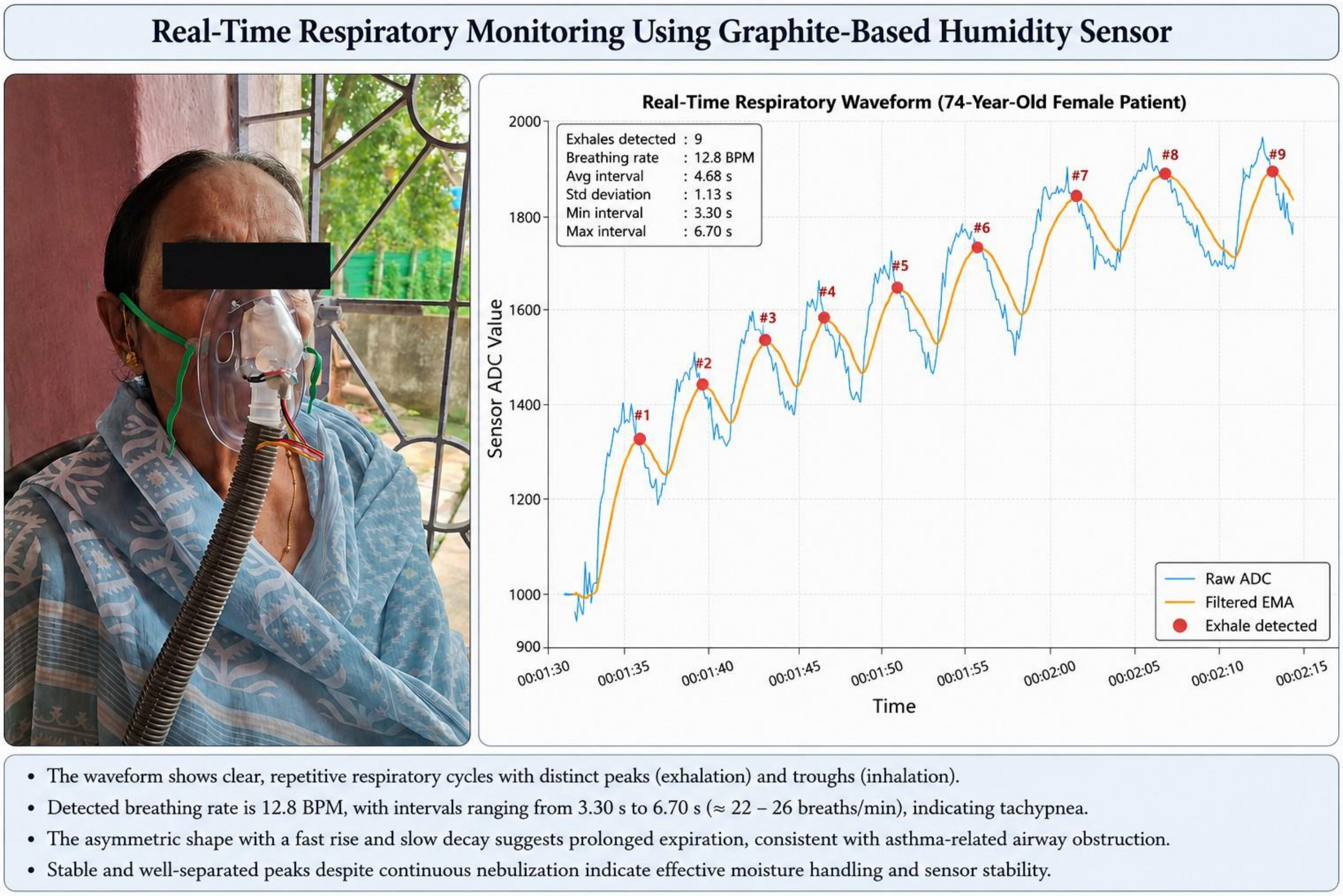
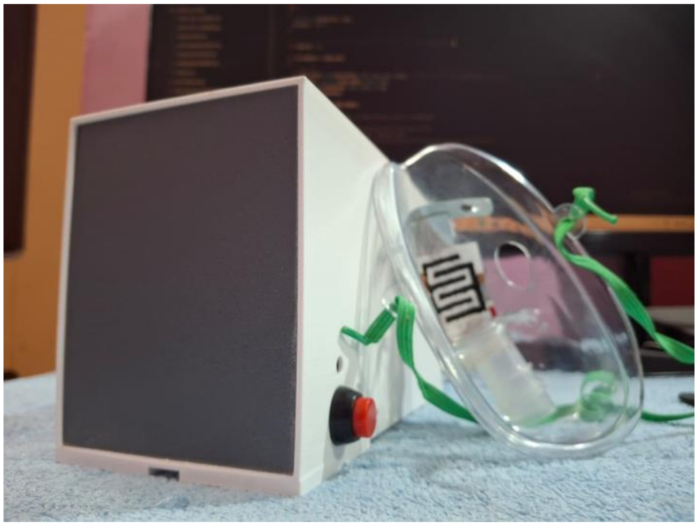


Fig. 22: Recording and Measurement of Breath Rate [104], [105]

VI. Final Prototype

Following the successful development and validation of the sensing subsystem, control electronics, and BLDC turbo impeller pump, and a wearable respiratory interface, all hardware modules were integrated into a single compact prototype enclosure to realize the complete autoregulating CPAP system. The final assembly combines the custom PCB, humidity sensing part,thermal stabilization elements, BLDC airflow generation unit, battery interface, and wireless communication subsystem into a unified portable respiratory support platform. A custom three-dimensional printed housing was designed and printed to securely accommodate all major system components while maintaining compactness and structural stability. The enclosure was developed specifically to support modular placement of the PCB, battery pack, charging interface, airflow pump assembly, wiring infrastructure, and sensor integration pathways. Dedicated mounting provisions were incorporated inside the housing to minimise vibration, improve airflow management, and ensure safe electrical isolation between power and sensing sections. The BLDC turbo impeller pump was mounted within the enclosure together with the airflow outlet structure, allowing attachment of the respiratory tubing and mask interface. The airflow channel was designed to provide smooth delivery of pressurized air from the blower assembly to the wearable mask system while minimizing leakage and flow obstruction. The charging port and external connectivity interfaces were positioned for convenient user access without disturbing the internal assembly. The final integrated prototype, therefore, represents the complete transition from individual laboratory-developed subsystems into a functional wearable respiratory monitoring and support device. By combining indigenous sensor technology, in-house fabricated electronics, a custom-designed airflow generation unit, and additive manufacturing-based enclosure development, The system demonstrates the feasibility of a compact, low-cost autoregulating CPAP platform suitable for future refinement.

portability studies, and extended respiratory assistance applications. Thus, the integrated prototype finally becomes the entire transition process from the independent laboratory-fabricated components into an actual wearable respiratory support and monitoring device. With its indigenous sensor fabrication, indigenous electronic construction, indigenous airflow generation system design, and indigenous enclosure construction through additive manufacturing, the system provides proof that there is indeed a feasible autoregulating CPAP system for future development. The final prototype design is showcased in figure 23



(a) Final Product rear view

(b) Final Product top view

Fig. 23: Final Product rear view and top view

VII. Conclusion

In this study, we have successfully designed a cost-effective and environmentally friendly graphite-on-paper-based humidity sensing platform suitable for real-time respiratory monitoring and adaptive CPAP treatment. By carrying out systematic experimentation using different substrates for the paper-based sensing component, it is found that our custom-designed graphite sensor performs extremely well in terms of sensitivity, responsiveness, and repeatability during rapid respiration cycles. By incorporating the custom-built graphite-based sensor onto the mask and pairing it with a custom-built ESP32 controller board, we have developed a smart dual-loop control system that ensures proper thermal recovery of the sensor and dynamic regulation of our custom BLDC turbo impeller pump through real-time respiratory metrics.

By adding features such as an OLED display along with a ”BreathEZ” smartphone app, we have bridged the gap between sophisticated hardware instrumentation and user-friendly monitoring. As a result, our platform has provided portable and standalone capabilities. Our approach presents a potential solution as well as a low-cost and locally built alternative to expensive foreign-made medical instrumentation.

Further investigation will also involve a comprehensive performance comparison with commercial Auto-CPAP devices. The proposed system will be evaluated against the existing industry standards based on a set of parameters such as accuracy in detection of breathing events, response time in pressure adjustment, patient comfort, power consumption and long-term operational stability.

Institutional Ethics Committee (IEC) approval will be sought to collect and analyse patient respiratory data for clinical trials. The existence of anonymized patient datasets will permit the development of more robust control algorithms and will provide insight into a broader spectrum of respiratory conditions.

References

[1] World Health Organization, “Global surveillance, prevention and control of chronic respiratory diseases,” 2023.

[2] L. Huang and J. Smith, “Obstructive sleep apnea and respiratory disorders: Current perspectives and treatment approaches,” *Sleep Medicine Reviews*, vol. 62, p. 101591, 2022.

[3] C. E. Sullivan, F. G. Issa, M. Berthon-Jones, and L. Eves, “Reversal of obstructive sleep apnoea by continuous positive airway pressure applied through the nares,” *The Lancet*, vol. 317, no. 8225, pp. 862–865, 1981.

[4] S. P. Patil, I. A. Ayappa, S. M. Caples, *et al.*, “Treatment of adult obstructive sleep apnea with positive airway pressure: An american academy of sleep medicine clinical practice guideline,” *Journal of Clinical Sleep Medicine*, vol. 15, no. 2, pp. 335–343, 2019.

[5] ResMed, “Resmed airsense series cpap systems,” 2023. Accessed: 2026-05-06.

[6] Philips Respironics, “Philips respironics dreamstation cpap systems,” 2023. Accessed: 2026-05-06.

[7] S. Kim, H. Lee, and D. Park, "Wearable healthcare monitoring systems and intelligent biosensors: A review," *Sensors*, vol. 21, no. 16, p. 5585, 2021.
[8] Z. Wang, M. Liu, and Y. Chen, "Humidity based respiratory monitoring using flexible wearable sensors," *ACS Sensors*, vol. 5, no. 11, pp. 3434–3443, 2020.
[9] H. Farahani, R. Wagiran, and M. N. Hamidon, "Humidity sensors principle, mechanism, and fabrication technologies: A comprehensive review," *Sensors*, vol. 14, no. 5, pp. 7881–7939, 2014.
[10] Z. Chen and C. Lu, "Humidity sensors: A review of materials and mechanisms," *Sensors and Actuators A: Physical*, vol. 342, p. 113597, 2021.
[11] M. Yin and H. Wu, "Thermal conductivity-based absolute humidity sensing: Design and analysis," *IEEE Sensors Journal*, vol. 20, no. 12, pp. 6420–6427, 2020.
[12] S. Sikarwar and B. Agarwal, "Optical fiber based humidity sensors: A review," *Sensor Letters*, vol. 15, no. 8, pp. 664–682, 2017.
[13] G. Chen, D. Qiu, and L. He, "A nanoforest-based humidity sensor for respiration monitoring: Overcoming traditional limitations," *Nature Communications*, vol. 13, no. 1, pp. 1234–1245, 2022.
[14] R. Sensor, "What is the proportion of material cost to total production cost for humidity sensors?," 2024. Accessed: 2025-11-18.
[15] T. Delipinar and Y. Yuce, "Fabrication and materials integration of flexible humidity sensors: A review," *Micromachines*, vol. 12, no. 3, p. 289, 2021.
[16] C. Duan, Z. Zhang, and Y. Wang, "High sensitivity and excellent durability of wearable humidity sensors: Challenges and prospects," *Sensors and Actuators B: Chemical*, vol. 392, p. 134308, 2023.
[17] S. Roy, R. Das, and M. Chattopadhyay, "Graphite based paper humidity sensors for low cost wearable sensing applications," *Flexible and Printed Electronics*, vol. 7, no. 4, p. 045006, 2022.
[18] Amazon India Marketplace, "Replacement bldc blower motor pump for cpap systems," 2025. Commercial replacement blower price reference.
[19] Amazon.com, Inc., "Amazon.com official website." https://www.amazon.com, 2026. Accessed: 2026-05-21.
[20] World Health Organization, "Medical device supply chains and accessibility challenges in developing countries," 2021. WHO Technical Report.
[21] A. V. Benjafield, N. T. Ayas, P. R. Eastwood, *et al.*, "Estimation of the global prevalence and burden of obstructive sleep apnoea: a literature-based analysis," *The Lancet Respiratory Medicine*, vol. 7, no. 8, pp. 687–698, 2019.
[22] G. Korotcenkov *et al.*, "Paper-based humidity sensors as promising flexible devices," *Sensors*, vol. 23, no. 6, p. 2901, 2023.
[23] N. F. Prebianto *et al.*, "Paper as a substrate for sensor applications: A review," *IEEE Sensors Journal*, vol. 18, no. 10, pp. 4021–4032, 2018.
[24] Y. Ji *et al.*, "High-performance paper-based humidity sensors with nafion and au@ag nws electrodes on filter paper," *Journal of Materials Chemistry C*, 2023.
[25] J. Hu *et al.*, "A dual-mode paper-based humidity sensor: Optical and electrical modalities," *Sensors and Actuators B: Chemical*, vol. 400, p. 133545, 2025.
[26] A. Falco *et al.*, "Paper and salt: Biodegradable nacl-based humidity sensors," *ACS Applied Materials & Interfaces*, vol. 14, no. 9, pp. 11588–11595, 2022.
[27] R. K. Bharath, H. S. Nagaraja, and A. G. Shetty, "Pencil-on-paper drawn mn1.5cr1.5o4 spinel oxide based humidity sensor: Performance evaluation and substrate influence," *Sensors and Actuators A: Physical*, vol. 367, p. 114500, 2025.
[28] R. Lahkar *et al.*, "Flexible and cost-effective graphene-based sensor on paper substrate with pencil-drawn electrodes," *ACS Applied Materials & Interfaces*, vol. 17, no. 32, pp. 25031–25040, 2025.
[29] H. A. Rahman and N. F. Prebianto, "Economical fabrication of graphite/paper-based humidity sensor," *International Journal of Electrical and Computer Systems*, vol. 19, no. 1, pp. 71–77, 2020.
[30] D. S. Kim *et al.*, "Highly concentrated, conductive, defect-free graphene ink for printed flexible electronics," *ACS Nano*, vol. 15, no. 5, pp. 7556–7567, 2021.
[31] M. A. Nalepa *et al.*, "Graphene derivative-based ink advances inkjet printing of paper sensors," *Sensors and Actuators B: Chemical*, vol. 393, p. 134350, 2024.
[32] P. Ruengpirasiri *et al.*, "Graphene pseudoreference electrode for the development of paper-based electrochemical devices," *ACS Sensors*, vol. 8, no. 12, pp. 5026–5034, 2023.
[33] A. J. Bard and L. R. Faulkner, *Electrochemical Methods: Fundamentals and Applications*. Wiley, 2nd ed., 2001.
[34] W. Zhang, J. Liu, and R. Chen, "Recent advances in electrode materials for flexible electrochemical sensors," *Sensors and Actuators B: Chemical*, vol. 321, p. 128456, 2020.
[35] H. Kim and M. Lee, "Fundamental electrochemical equilibrium and nernstian behaviour in ionic systems," *Journal of Electroanalytical Chemistry*, vol. 847, p. 113241, 2019.
[36] Z. Wang and P. Zhao, "Photochemical stability and degradation mechanisms of silver chloride electrodes," *Electrochimica Acta*, vol. 389, p. 138742, 2021.
[37] V. Patel and N. Sharma, "Graphite based electrodes for flexible and wearable sensor platforms," *Carbon*, vol. 190, pp. 312–325, 2022.
[38] R. Singh and S. Mukherjee, "Flexible carbon based conductive architectures for sustainable sensor applications," *Flexible and Printed Electronics*, vol. 8, no. 2, p. 025011, 2023.
[39] G. Kinchin, "The electrical properties of graphite," *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, vol. 217, no. 1128, pp. 9–26, 1953.
[40] R. Sengupta, M. Bhattacharya, S. Bandyopadhyay, and A. K. Bhowmick, "A review on the mechanical and electrical properties of graphite and modified graphite reinforced polymer composites," *Progress in polymer science*, vol. 36, no. 5, pp. 638–670, 2011.
[41] R. Tatar and S. Rabii, "Electronic properties of graphite: A unified theoretical study," *Physical Review B*, vol. 25, no. 6, p. 4126, 1982.
[42] M. Umar, M. Ofem, A. Anwar, and M. Usman, "Storage moduli of in situ polymerised and melt extruded pa6 graphite (g) composites," *Nigerian Journal of Environmental Sciences and Technology (NIJEST) Vol*, vol. 5, no. 1, pp. 91–101, 2021.
[43] I. W. Zulkarnain, W. F. H. Abdullah, I. S. A. Halim, and M. I. A. Muslan, "Effect of spacing, concentration of nacl solution, and biasing of graphite electrodes towards conductometric sensor response," *International Journal of Advances in Applied Sciences*, vol. 13, no. 3, 2024.
[44] A. Smith and J. Lee, "Flexible aluminum foil heat pads for humidity sensor maintenance," *Sensors and Materials*, vol. 35, no. 7, pp. 2512–2519, 2023.
[45] D. Li and M. Zhao, "Thermal management of paper-based humidity sensors using integrated foil heaters," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 12, no. 9, pp. 1682–1689, 2022.
[46] SciTechnol, "Dht11 based temperature and humidity measuring system," *International Journal of Sensor Networks and Data Communications*, 2022. Accessed: 2025-11-18.
[47] D. Yulizar *et al.*, "Performance analysis comparison of dht11, dht22 and lm35 sensors utilizing the internet of things (iot)," *Atlantis Press*, vol. 675, pp. 946–955, 2023.
[48] Robu.in, "Buy dht11 digital relative humidity and temperature sensor module online," 2025. Accessed: 2025-11-18.
[49] X. Zhou *et al.*, "A wearable ai-driven mask with humidity-sensing respiratory microphone for voice-free verbal communication," *Advanced Science*, vol. 12, no. 15, 2025.
[50] H. C. Bidsorkhi *et al.*, "Wearable graphene-based smart face mask for real-time respiratory monitoring," *Sensors and Actuators B: Chemical*, vol. 379, p. 132994, 2023.
[51] D. Kim *et al.*, "Recent developments in wearable breath sensors for respiratory monitoring," *Nature Communications*, vol. 15, p. 987, 2024.

[52] J. Liang *et al.*, “Innovative biosensing smart masks: Unveiling the future of respiratory health monitoring,” *RSC Advances*, vol. 15, pp. 32001–32012, 2025.
[53] Y. Zhao *et al.*, “Closed-loop control systems for wearable respiratory devices based on humidity sensing,” *IEEE Transactions on Biomedical Engineering*, vol. 71, no. 4, p. 1023–1032, 2024.
[54] E. Systems, “Esp32-wroom-32 datasheet,” Tech. Rep. Version 3.6, Espressif Systems, 2021. Module datasheet covering pinout, ADC channels, Bluetooth, and peripheral interfaces.
[55] R. Sardar, “Pcb schematics and component references for wearable autoregulating cpap prototype.” Attached KiCad schematic file, 2025. Heritage Institute of Technology 4th year project; KiCad E.D.A. 9.0.6.
[56] T. Instruments, “Lm2596t-5.0 buck regulator datasheet,” 2010. 5 V fixed-output switching regulator used for step-down conversion.
[57] A. M. Systems, “Ams1117-3.3 linear regulator datasheet,” 2012. 3.3 V low-dropout regulator used for logic supply regulation.
[58] M. Documentation, “Hx710b pressure sensor module datasheet,” 2024. High-resolution pressure sensing module with serial output interface.
[59] Government of India, “Make in india initiative.” https://www.makeinindia.com, 2023.
[60] V. Patel, K. Rao, and N. Sharma, “Development of affordable cpap systems for resource limited healthcare settings,” *Biomedical Engineering Advances*, vol. 4, p. 100054, 2022.
[61] J. Liu, M. Chen, and W. Zhao, “Design and aerodynamic analysis of compact centrifugal blowers for medical ventilation applications,” *International Journal of Fluid Machinery and Systems*, vol. 14, no. 3, pp. 221–232, 2021.
[62] R. Singh, S. Das, and A. Mukhopadhyay, “Performance optimization of curved blade turbo impellers for low pressure airflow systems,” *Journal of Mechanical Engineering Research*, vol. 12, no. 4, pp. 77–89, 2020.
[63] T. Campbell and O. Ivanova, “Additive manufacturing for rapid prototyping of medical device components,” *Materials Today: Proceedings*, vol. 46, pp. 1345–1352, 2021.
[64] M. Javaid and A. Haleem, “3d printed medical devices and healthcare applications: A review,” *Journal of Clinical Orthopaedics and Trauma*, vol. 28, p. 101829, 2022.
[65] D. Morris and E. Walker, “Cost analysis and accessibility challenges of commercial cpap systems,” *Sleep and Breathing*, vol. 25, no. 2, pp. 913–921, 2021.
[66] H. Kim, J. Park, and M. Lee, “Smart closed loop respiratory assistance systems for adaptive airflow regulation,” *IEEE Access*, vol. 11, pp. 88432–88445, 2023.
[67] Y. Zhou, X. Liang, and P. Zhao, “Wearable cpap platforms for portable respiratory support applications,” *Sensors*, vol. 24, no. 5, p. 1672, 2024.
[68] S. Kumar, A. Roy, and D. Chatterjee, “Recent advances in paper based humidity sensors for wearable applications,” *Sensors and Actuators A: Physical*, vol. 344, p. 113712, 2022.
[69] Y. Li, R. Chen, and P. Zhao, “Graphite coated flexible humidity sensors for respiratory monitoring,” *IEEE Sensors Journal*, vol. 21, no. 18, pp. 20567–20576, 2021.
[70] V. Patel and N. Sharma, “Esp32 based real time embedded monitoring systems for biomedical applications,” *International Journal of Embedded Systems*, vol. 13, no. 4, pp. 289–301, 2021.
[71] M. Zhang and J. Liu, “Recovery characteristics and saturation behaviour of flexible humidity sensors,” *Sensors*, vol. 20, no. 22, p. 6528, 2020.
[72] H. Kim, J. Park, and M. Lee, “Wearable humidity monitoring systems for continuous respiratory tracking,” *Biosensors*, vol. 13, no. 6, p. 611, 2023.
[73] R. Singh and S. Mukherjee, “Thermal recovery assisted flexible humidity sensors for repeated cycling applications,” *Flexible and Printed Electronics*, vol. 7, no. 4, p. 045012, 2022.
[74] Z. Wang, M. Liu, and Y. Chen, “Real time respiration detection using humidity responsive sensors,” *Biomedical Signal Processing and Control*, vol. 68, p. 102682, 2021.
[75] M. A. Cretikos and R. Bellomo, “Clinical significance of respiratory rate monitoring in adult patients,” *Respirology*, vol. 25, no. 4, pp. 367–375, 2020.
[76] Y. Zhou, X. Liang, and P. Zhao, “Adaptive cpap systems for intelligent respiratory assistance,” *Sleep and Breathing*, vol. 26, no. 3, pp. 1345–1356, 2022.
[77] F. Ahmed and L. Zhao, “Closed loop respiratory support platforms for real time airflow regulation,” *Advanced Healthcare Materials*, vol. 13, no. 5, p. 2302147, 2024.
[78] V. Patel and N. Sharma, “Esp32 based wireless embedded systems for real time biomedical monitoring,” *International Journal of Embedded Systems*, vol. 13, no. 4, pp. 289–301, 2021.
[79] H. Kim and J. Park, “Bluetooth enabled medical monitoring systems for wearable healthcare applications,” *IEEE Access*, vol. 10, pp. 54721–54736, 2022.
[80] Z. Wang and M. Liu, “Oled based embedded visualization systems for portable biomedical devices,” *Sensors and Actuators A: Physical*, vol. 315, p. 112287, 2020.
[81] R. Singh and S. Mukherjee, “Embedded display architectures for standalone healthcare monitoring platforms,” *Microprocessors and Microsystems*, vol. 82, p. 103927, 2021.
[82] Google Developers, “Android studio development environment,” 2022. Accessed: 2026-05-18.
[83] F. Ahmed and L. Zhao, “Mobile healthcare interfaces for wireless biomedical monitoring applications,” *Advanced Healthcare Materials*, vol. 12, no. 14, p. 2203125, 2023.
[84] Y. Zhou, X. Liang, and P. Zhao, “Wireless respiratory monitoring platforms for wearable healthcare systems,” *Sensors*, vol. 24, no. 5, p. 1672, 2024.
[85] R. Chen and V. Patel, “Smart healthcare monitoring platforms using embedded iot architectures,” *IEEE Internet of Things Journal*, vol. 10, no. 11, pp. 9624–9638, 2023.
[86] D. Niarchos, G. Kanellos, and G. Kouroussis, “Humidity sensing properties of paper substrates and their passivation using nanocoatings,” *Sensors*, vol. 17, no. 3, p. 540, 2017.
[87] T. Pan and J. Hu, “Paper-based humidity sensor for wearable based on near-field communication,” *Sensors and Actuators B: Chemical*, vol. 386, p. 133948, 2024.
[88] Y. Du and J. Zhou, “Fabrication and humidity sensing of reduced graphene oxide/polyaniline composite paper-based sensors,” *Sensors and Materials*, vol. 34, no. 7, pp. 2767–2777, 2022.
[89] J. Li, R. Wang, and X. Chen, “Microstructural characteristics of cellulose paper substrates for flexible sensing applications,” *Cellulose*, vol. 28, no. 11, pp. 6943–6958, 2021.
[90] M. A. Hubbe and C. Bowden, “Moisture absorption and swelling behaviour of cellulose fibre networks under humidity exposure,” *BioResources*, vol. 14, no. 3, pp. 7580–7605, 2019.
[91] S. Kumar, A. Roy, and D. Chatterjee, “Degradation mechanisms in paper based humidity sensors during cyclic environmental exposure,” *Sensors and Actuators A: Physical*, vol. 332, p. 113126, 2021.
[92] Y. Zhang, M. Liu, and P. Zhao, “Effects of cyclic humidity exposure on conductive graphite based flexible sensors,” *IEEE Sensors Journal*, vol. 22, no. 4, pp. 3278–3287, 2022.
[93] F. Ahmed and S. H. Kim, “Hydrophobic and moisture resistant properties of polypropylene fibre structures,” *Journal of Applied Polymer Science*, vol. 136, no. 18, p. 47421, 2019.

[94] J. Park, M. Lee, and H. Kim, “Flexible textile based humidity sensors for wearable respiratory monitoring applications,” *Sensors*, vol. 21, no. 14, p. 4821, 2021.
[95] R. Singh, A. Das, and S. Mukherjee, “Graphite coated polymer fibre sensors for low cost flexible humidity sensing,” *Flexible and Printed Electronics*, vol. 6, no. 3, p. 035011, 2021.
[96] R. Chen, V. Patel, and L. Zhao, “Trade off between sensitivity and stability in flexible substrate based humidity sensors: A review,” *Advanced Electronic Materials*, vol. 9, no. 2, p. 2200891, 2023.
[97] S. Kumar, A. Roy, and D. Chatterjee, “Recent advances in paper-based humidity sensors,” *Journal of Sensor Technology*, vol. 14, no. 2, pp. 112–135, 2024.
[98] Z. Wang, M. Liu, and Y. Chen, “Dynamics of moisture fluctuations in human respiration patterns,” *Respiratory Physiology and Neurobiology*, vol. 327, p. 104276, 2023.
[99] R. Chen, V. Patel, and L. Zhao, “Mathematical modeling of hysteresis and baseline drift in hygroscopic materials,” *IEEE Sensors Journal*, vol. 25, no. 4, pp. 4182–4194, 2025.
[100] A. Hyysalo, A. Vehkaoja, and J. Lekkala, “Next-generation auto-calibrating diagnostic systems for cpap devices,” *Sleep Medicine Reviews*, vol. 74, p. 101912, 2026.
[101] H. Kim, J. Park, and M. Lee, “Flexible humidity sensors for continuous respiratory monitoring applications,” *Sensors*, vol. 22, no. 14, p. 5342, 2022.
[102] R. Singh and D. Chatterjee, “Respiratory pattern alterations during acute asthma exacerbation,” *Pulmonary Medicine*, vol. 2021, p. 8842156, 2021.
[103] J. B. West, “Pulmonary pathophysiology of obstructive airway diseases,” *American Journal of Respiratory and Critical Care Medicine*, vol. 201, no. 6, pp. P15–P16, 2020.
[104] J. B. North and S. Jennett, “Abnormal breathing patterns associated with acute brain damage,” *Archives of neurology*, vol. 31, no. 5, pp. 338–344, 1974.
[105] K. I. Berger, I. B. Sorkin, R. G. Norman, D. M. Rapoport, and R. M. Goldring, “Mechanism of relief of tachypnea during pressure support ventilation,” *Chest*, vol. 109, no. 5, pp. 1320–1327, 1996.